CH4　材料組成律

4-1. 材料拉伸試驗

一、材料的機械性質及虎克定律

1. **材料的機械性質及虎克定律**：使用標準試片，材料在拉伸作用力下，所表現出來的變形和破壞等方面的機械特性，可表現在應力-應變圖中，如圖所示，分析如下：

(1) 比例限與彈性限：

A. 在拉伸（或壓縮）的初始階段應力σ與應變爲直線關係，如圖中對應的應力值σ_{pl}稱爲比例極限，亦即比例限以下之應力值與應變成正比，表示爲$\sigma = E\varepsilon$。

B. 應力-應變曲線上當應力增加到某一應力值時，再將應力降爲零，則應變隨之消失；一旦超過該應力值，卸載後，有一部分應變不能消除，則此應力值定義爲彈性限。

(2) 材料的降伏(yielding)與破壞：

A. 當受力超過某一應力值時，應力與應變不再成正比，當應力超過該應力值時，在應力增加很少或不增加時，應變會很快增加，這種現象叫降伏，則該應力值稱之爲降伏應力。

B. 材料經過降伏階段以後，因塑性變形使其組織結構得到調整，若需要增加應變則需要增加應力，曲線又開始上升，此過程稱爲材料硬化過程，到最高點的強度是材料能承受的極限應力σ_u。

C. 當拉伸到強度極限時，在試件的某一局部範圍內橫截面急劇縮小，曲線開始下降，此過程稱爲材料縮頸過程，形成縮頸現象。

斷面收縮率：$\delta = \dfrac{l_1 - l_0}{l_0} \times 100\%$

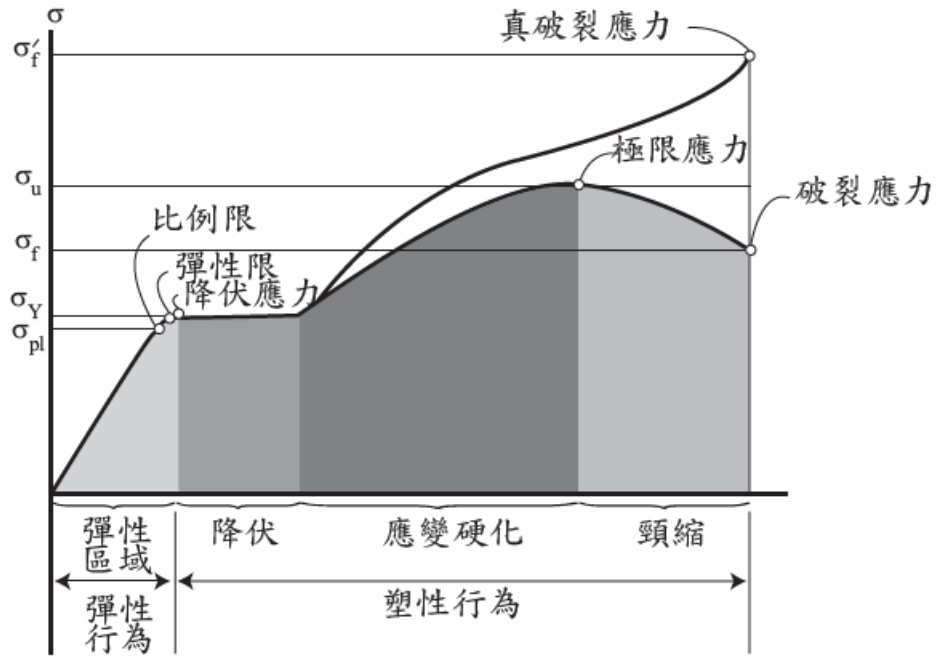
伸長率：$\delta = \dfrac{l_1 - l_0}{l_0} \times 100\%$

(3)虎克定律

A.對大部份材料而言，皆具有這種線彈性之性質，我們稱該材料爲線彈性體，在線彈性範圍下，當應力不超過比例極限時，則正應力與縱向線應變成正比 $\Rightarrow \sigma = E \cdot \varepsilon$，即是著名的虎克定律(Hooke's law)，表示成 $\sigma = E\varepsilon$，其中E爲彈性係數(modulus of elasticity)，又稱爲楊氏模數(Young's modulus)，爲材料之常數，其單位與應力相同，在SI 制爲Pa，而英制則爲psi，彈性模數爲應力應變曲線中線彈性區中斜直線之斜率，對於同一種材料其值爲定值，不會因製程不同而改變。

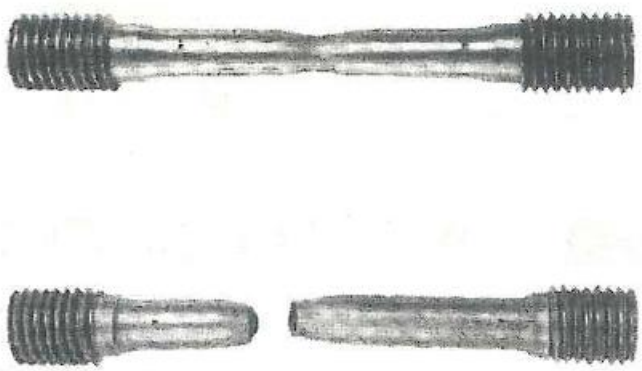
B.當桿件受力，橫截面上的正應力不超過比例極限時，桿件的伸長量 δ 與軸力P及桿原長L成正比，與橫截面面積A成反比，即 $\delta \propto \frac{PL}{A}$，引入比例常數E，則可寫爲 $\delta = \frac{PL}{EA} \Rightarrow$ 虎克定律。

C.材料的彈性模數是經由實驗測定出來的，彈性模數表示在受拉（壓）時，材料抵抗彈性變形的能力，EA越大，桿件的變形就越小，故稱EA爲桿件抗拉（壓）剛度。



【說明】：

1. 彈性與線性二名詞意義不同。彈性意指外力去除後，其應變是否能沿原路徑回復；線性意指回復路徑之型態。若材料加載之後產生應變，而在外力去除之後卻不循原路徑回復，亦即不能回復原狀者。
2. 延性材料試件會造成頸縮現象



二、應力應變圖之機械性質

(一)應變偏距法

大部分的金屬材料，彈性變形僅存在應變的 0.005 之內變形超過此點:應力不再正比於應變(虎克定律不適用)⇒永久且不能回復的塑性變形(Plastic Deformation)，但有些金屬降伏強度並不明顯，可採用降伏強度(σy)由 0.002 (0.2%)應變偏距法來決定:

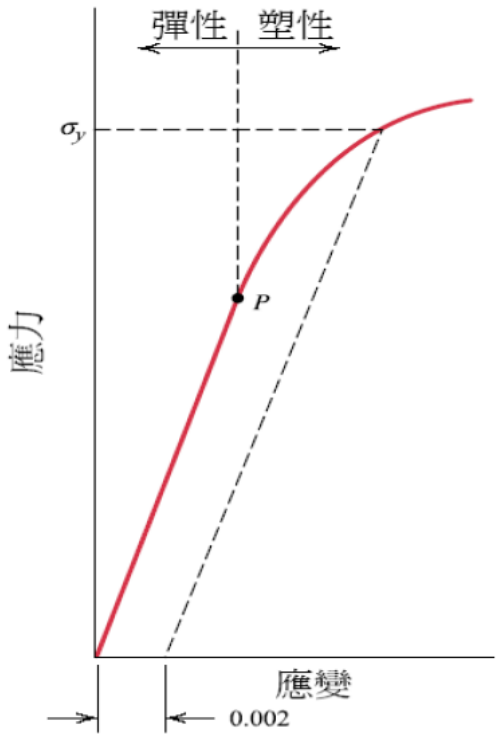
步驟:

1. 找到 0.0022 的位置
2. 劃與線性曲線平行線與應

力-應變曲線交叉

結果:

- 交叉點為降伏強度
- 預估彈性轉塑性區 P 點



【說明】：

1. 脆性材料超過比例限一點點就會斷裂，不會有明顯的變形，亦無明顯的頸縮。
2. 材料極限降伏強度與破裂強度一致，破裂應變遠小於延性材料之破裂應變。
3. 桿件破壞型態如圖所示，斷面切齊。



(二)應力應變圖之機械性質

1. 材料均質性(homogeneous)及等向性(isotropic)

   某一物體之材料其內部任一點，全都具有相同之性質者，稱為均質材料(homogeneous material)，均質材料在各方向中之性質並不需要相同，例如在軸向與側向之彈性性質，即有可能不同；而材料各個方向之力學性質均相同時，稱為具材料等向性。對於一均質且等向之材料，在線彈性範圍內，側向應變與軸向應變成正比，亦即帕桑比為一常數，很多工程材料均具有此種特性。因此，一物體中之材料，其內部各點之間，並未具有相同之性質者，稱為非均質材料，常見之非均質材料，例如土壤材料，複合材料等。等向性、均質之線彈性材料，其力學性質只需兩個係數就可充分描寫，即彈性係數 E 及剪力彈性係數 G 或帕桑比值(Poisson

ratio)v。一般而言金屬材料常具等向性，木材、竹子因纖維紋路存在，縱向、橫向力學性質迥異，不具等向性之材料稱為非等向性材料，含節理之岩石、頁岩以及土壤均為非等向性材料。

2. 延性（ductility）與脆性（brittle）

   延性（ductility）是另一重要的機械性質，它是測量持續至破裂時所能承受塑性變形之程度，材料在破裂時僅有非常小或甚至沒有塑性變形，則稱為脆性（brittle），破裂應變小於 5%為脆性材料。

3. 延展比(ductility ratio)」作為判別延性材料或脆性材料的依據。以 μ 表示延展比，其定義為 $\mu = \frac{\varepsilon_u(\text{材料破壞時之應變})}{\varepsilon_y(\text{材料降伏之應變})}$ 。

4. 潛變（Creep）

   若於材料上緩慢施加一 P 力，於 t_1 時間完成之伸長量或變形；當 $t > t_1$ ，若載重持續不變，但變形卻隨時間增加而增加，這種在長時間作用下，載重不變，變形卻增加的現象，稱為潛變（Creep），例如混凝土及預力混凝土構件需考慮此項影響。

5. 鬆弛（relaxation）

   當有初始應力 存在於材料上，經過長時間作用，即使支承端並未移動， 亦隨時間增加而造成下降，最終達成定值，稱為鬆弛（relaxation）。

6. 應變能

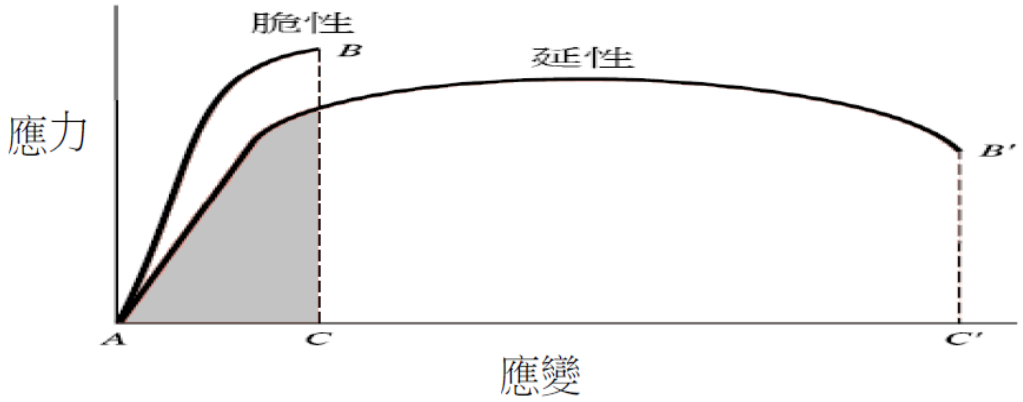
   在應力－應變曲線下的面積代表每單位體積的材料所能吸收的能量，稱之為應變能。

7. 回彈模數（modulus of resilience）

   即在降伏點之前，應力與應變圖下所圍之面積。回彈模數愈大，即表示在降伏之前材料所儲存的能量愈多。

8. 韌性（toughness）

   韌性（toughness）是測量材料到斷裂時所能吸收的能量，破裂韌性是指當有裂縫存在時，材料抵抗破裂的性質，對於靜態（低應變速率）狀況，韌性可藉由拉伸的應力對應變的測試結果來得到，它是指 σ －E 曲線到斷裂點以下所圍的面積。由圖中之兩種典型材料的應力對應變曲線來看，即使脆性材料具有高降伏和拉伸強度，但由於缺乏延性，所以其韌性仍較延性材料低，這可由比較圖中 A B C 和 A B’ C’ 面積來得到。



(三)**真應力應與真應變圖**

當應力通過拉伸強度，持續變形所需的應力會降低，此似乎意指金屬變弱了。事實上情況並非皆如此，而且其強度反而是增加，只是由於變形發生時頸縮區內橫截面面積快速的減小，反而降低了試片的負荷能力。在應力應變圖中，如果考慮材料在進入塑性階段時，因荷重的增加而產生面積和長度的改變，有時使用真應力－真應變來說明會更有意義，真應力定義成負荷 F 除以變形發生時的瞬間橫截面積 Ai（即通過拉伸點之頸縮）

$$\text{真應力}\sigma_{\mathrm{T}} = \frac{\mathrm{F}}{\mathrm{A}_i}$$

$$\text{真應變}\varepsilon_{\mathrm{T}} = \ln(\frac{\mathrm{l}_0}{\mathrm{l}_i})$$

假如在變形期間沒有體積變化 $\mathrm{A}_0 l_0 = A_i l_i$

真應力、真應變與工程應力及工程應變間的關係為

$$\sigma_{\mathrm{T}} = \sigma(1+\varepsilon)$$

$$\varepsilon_{\mathrm{T}} = \ln(1+\varepsilon)$$

三、 應變能密度及應變能

(一)**單軸應力之應變能**

應變能(strain energy)當負載作用於物體上，變形將產生，假設能量不會損失減少，由負載作用的的外功將會轉成內功稱為應變能(strain energy)，此能量能永遠保持正值儲存於物體內，是以正應力或是剪應力所造成，如圖所示一體積元素若受正向應力 σ_z 則產生頂面及底面的力 $dF = \sigma_z dA = \sigma_z dxdy$ ，若此力逐漸增加作用於體積元素上，其值由零增加到 dF ，體積元素的位移 $d\delta = \varepsilon_z dz$ ，則 dF 做功:

$$dU = \frac{1}{2} dF \bullet d\delta = \frac{1}{2}\sigma_z dxdy \bullet \varepsilon_z dz$$

$$\Rightarrow dU = \frac{1}{2}\sigma_z \varepsilon_z dxdydz = \int u d\forall$$

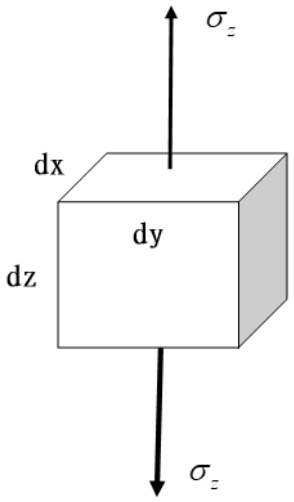
其中 體積元素之體積 $d\forall = dxdydz$ 、u 表應變能密度

若物體承受單軸向應力

$$\text{應變能U} = \int \frac{\sigma\varepsilon}{2} d\forall$$

若在材料線彈性下由虎克定律可將應變能表示為:

$$\text{應變能U} = \int \frac{\sigma^2}{2\text{E}} d\forall$$



(二)**剪應力的應變能**

爲了分析的方便，從受剪切桿中截取如圖所示的體積元素，該體積元素處於純剪切應力狀態，假想其在一個面（如左側面）上被固定起來，則在剪應力由零逐漸增加到 τ 值的過程中，體積元素將發生如圖所示的變形，與此對應的剪應變由零增加到 γ 值，其右側面向下的位移爲 $\Delta = \gamma dx$。當材料在線彈性範圍內工作時，其 τ 與 γ 成正比（圖 b ），應力元素各表面上的剪力在體積元素變形過程中所作的功爲

$$d\text{U} = \frac{1}{2}(\tau dydz)\cdot\Delta = \frac{1}{2}(\tau dydz)(\gamma dx)$$

上式中，作功的力是單元體右側面上的剪力。由於剪應變 γ 很小，其餘各面上的剪力，在其作用方向上沒有位移，都沒有在其作用方向上作功。故應力元素內積蓄的應變能爲

$$d\text{U} = dW = \frac{1}{2}\tau\gamma\cdot d\forall$$

應力元素內積蓄的應變能密度則爲

$$u = \frac{dU}{d\forall} = \frac{1}{2}\tau\gamma$$

這表明u等於$\tau - \gamma$直線下的面積，由虎克定律$\tau = G\gamma$，應變能密度又可以寫成下列形式

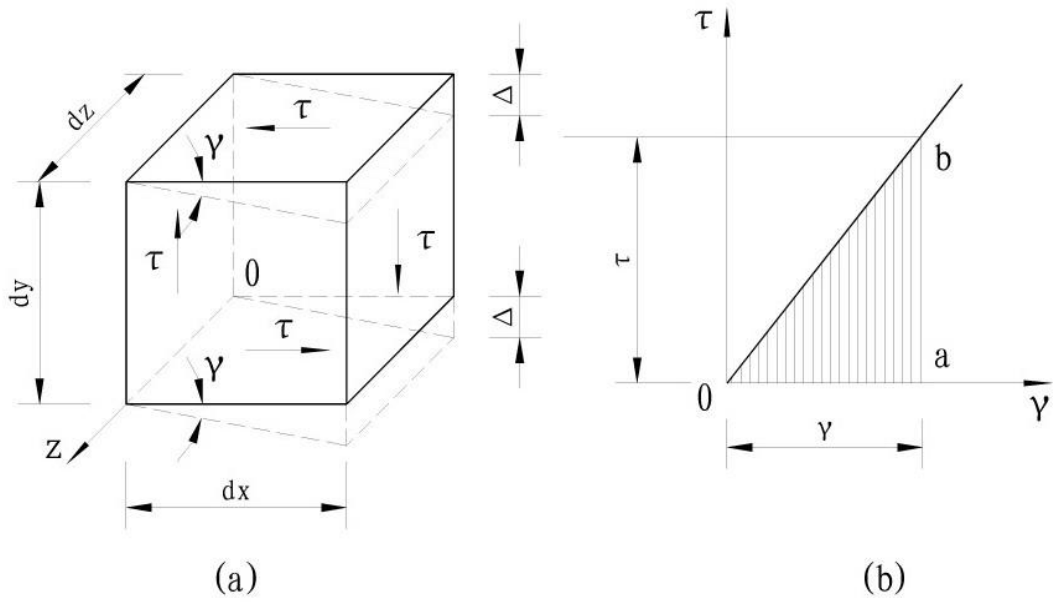
$$u = \frac{1}{2}\tau\gamma = \frac{\tau^2}{2G} = \frac{G\gamma^2}{2}$$

其中 體積元素之體積$d\forall = dxdydz$、u 表應變能密度

應變能$U = \int \frac{\tau\gamma}{2} d\forall = \int u d\forall$

若在材料線彈性下由虎克定律可將應變能表示為:

應變能$U = \int \frac{\tau^2}{2G} d\forall$



(a) (b)

(三)多軸應力之應變能

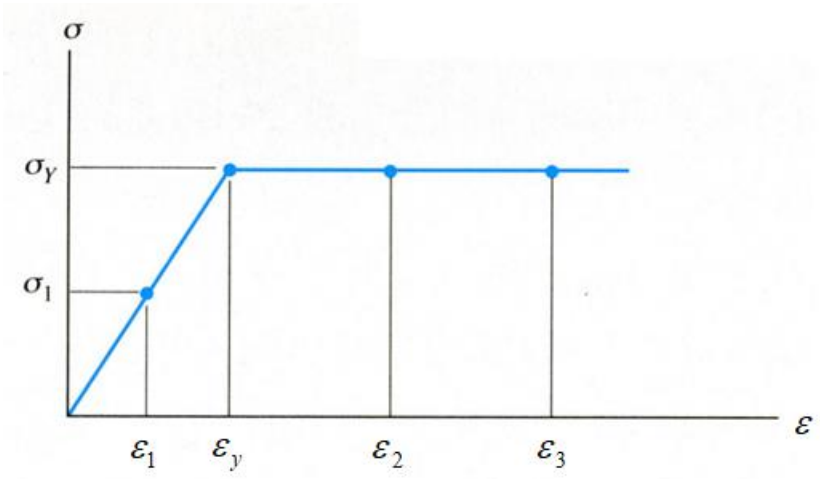
多軸應力之應變能可經由前面推導更推廣為:

$$\text{應變能}U = \int (\frac{\sigma_x\varepsilon_x}{2} + \frac{\sigma_y\varepsilon_y}{2} + \frac{\sigma_z\varepsilon_z}{2} + \frac{\tau_{xy}\gamma_{xy}}{2} + \frac{\tau_{xz}\gamma_{xz}}{2} + \frac{\tau_{yz}\gamma_{yz}}{2}) d\forall$$

4-2. 材料非彈性軸向變形

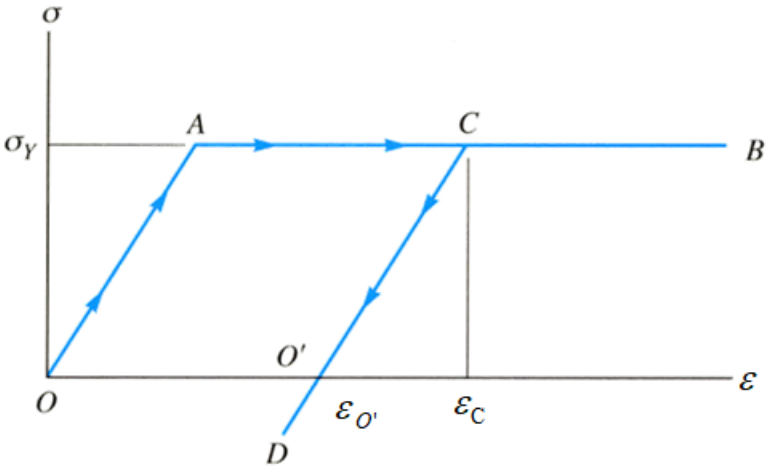
一、材料彈性-完全塑性 (elastic-perfectly plastic)

構件設計成負載後會導致材料降伏而產生永久變形。此種構件通常使用具有高延性的金屬如退火低碳鋼，其應力 - 應變曲線可模型化如圖所示。具有此種性質的材料通常稱為彈性-完全塑性 (elastic-perfectly plastic) 或彈塑 (elastoplastic)



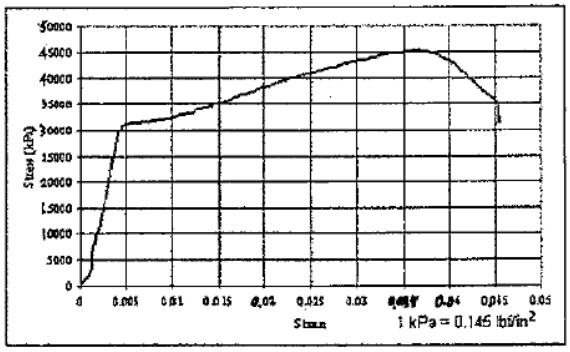
二、殘留應力 (residual stress)

若一軸向負載構件或一組構件為靜不定系統而可承受拉力或壓力，在受到過度負載時會導致材料降伏，當負載移除後會在構件產生殘留應力 (residual stress)。產生殘留應力的理由是於卸載後材料的彈性恢復所致。欲解此種問題，可將構件的負載及卸載過程考慮成正值負載 (加載) 及負值負載 (卸載) 的重疊。在負載過程由 O 至 C，產生塑性應力分佈，而在卸載過程 CD，只有彈性應力分佈。



範例 1

The followings are the uniaxial tensile testing results for a given specimen. (x-axis ranges from 0 – 0.05; y-axis ranges from 0 – 50000 kPa in case if the printing is not clear) Please answer the following questions.



(a) Please estimate the Young's modulus and yield strength of the structure. (4 Pts)

(b) Please determine the toughness of the material. (4 Pts)

(c) Please sketch the essential true stress-strain curve. (4 Pts)

(d) If this specimen can be idealized as a uniaxial geometry with a diameter of 1 cm and a length of 10 cm, please estimate the axial stiffness of this specimen. (4 Pts)

(e) (nothing related to part (a)-(d)) A certain isotropic material yields in uniaxial tension at the stress of 140 MPa. At a certain point in a structure made of this material, two of the three principal stresses are 20MPa tensile and 90 MPa compressive. If there is to be no yielding, what is the allowable range of the third principal stress according to the Tresca criterion? (9 Pts)

【105 成大機械】

【解析】

(a)

$$\mathrm{E}=\frac{\sigma}{\varepsilon}=\frac{30000}{0.004}=7.5\times10^{6}(KPa)=7.5(GPa)$$

$$\sigma_{Y}\cong 32000\,(KPa)=32(MPa)$$

(b)

韌性（toughness）是測量材料到斷裂時所能吸收的能量，σ－E曲線到斷裂點以下所圍的面積

$$(3+48+14)\times 25=1625\,(KPa)$$

(d)

$$\delta=\frac{PL}{EA}\Rightarrow P=\frac{EA}{L}\delta\Rightarrow K=\frac{EA}{L}=\frac{7.5\times10^{3}\times2\times5^{2}}{100}=5890.5(N/mm)$$

(e)

$\sigma_1=20(MPa)$ 、 $\sigma_2=-90(MPa)$

由最大剪應力理論

$$\tau_{\max} \le \frac{\sigma_Y}{2} \le 70(MPa)$$

$$\frac{20-\sigma_3}{2} \le 70(MPa) \Rightarrow \sigma_3 \ge -120(MPa)$$

$$\frac{90+\sigma_3}{2} \le 70(MPa) \Rightarrow \sigma_3 \le 50(MPa)$$

$$120(MPa) \le \sigma_3 \le 50(MPa)$$

範例 2

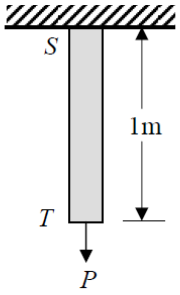
直徑 5 mm 之圓桿 ST 受軸向力 P 作用如下圖(a)。圓桿材料的楊氏係數為 2×10^5 MPa，其應力(σ)-應變(ε)曲線如下圖(b)。

㈠圖(b)的 σ-ε 曲線可透過何種試驗獲得？曲線上相應於 A 點的應力在力學上的名稱為何？（4 分）

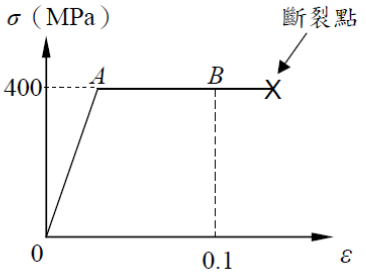
㈡ ST 可反抗的最大施力 P_{max} 為多少？如控制 P 使其自零慢慢增加至 P_{max}，試述 ST 變形的過程及其最終的狀態。（10 分）

㈢如施加適當負載將圓桿材料的應力-應變狀態帶到 B 點，則圓桿中此時的應變能密度為多少？（5 分）

㈣如到 B 點後，將負載完全撤去，則圓桿最後的伸長量為多少？（6 分）



圖(a)



圖(b)（本圖為示意圖，未按比例繪畫）

【105 機械高考】

【解析】

(1)拉伸試驗(tensile test)、降伏應力

(2) $P_{\max}=\sigma_y\times\dfrac{\pi d^2}{4}=400\times\dfrac{\pi\times5^2}{4}=7854(N)$

OA 過程：

隨著負載增加桿件長度伸長

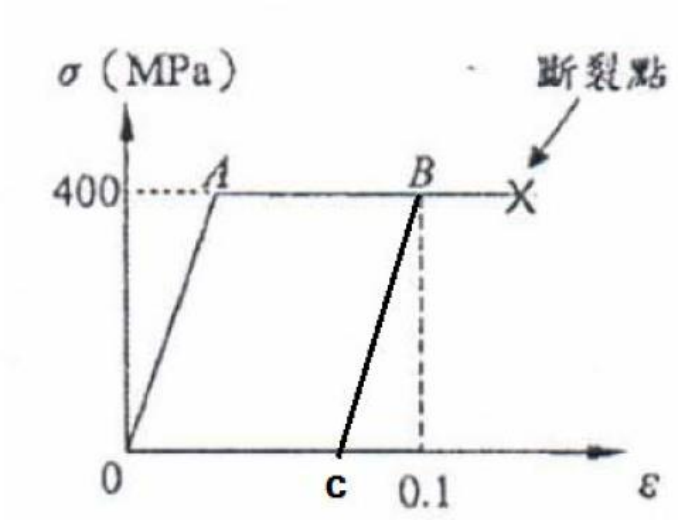
AB 過程：

過 A 點進入降伏區，降伏區負載保持定值，桿件長度持續伸長至斷裂。

(3)應變能密度即 OAB 線下面積

A 點位置 $\varepsilon_y=\dfrac{\sigma_y}{E}=\dfrac{400}{2\times10^5}=0.002$

$$u=\frac{1}{2}(0.1+0.098)\times400=39.6(\frac{MJ}{m^3})$$



考慮 B 點卸載至 C 點(且 CB 與 OA 具相同斜率)

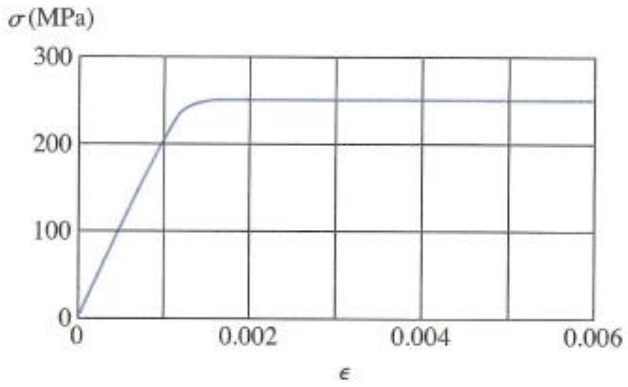
$\varepsilon_b = 0.1 \quad \varepsilon_y = 0.002$

$\varepsilon_C = \varepsilon_b - \varepsilon_y = 0.1 - 0.002 = 0.098$

得最後伸長量 $\delta_C = L \times \varepsilon_C = 1000 \times 0.098 = 98(mm)$

範例 3

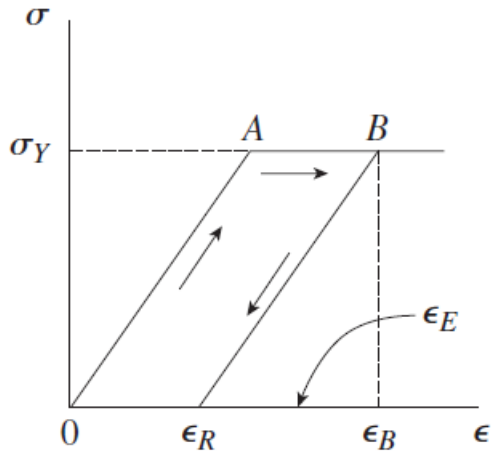
A bar of length 0.8 m is made of a structural steel having the stress-strain diagram shown in the figure. The yield stress of the steel is 250 MPa and the slope of the initial linear part of the stress-strain curve (modulus of elasticity) is 200 GPa. The bar is loaded axially until it elongates 2.5 mm, and then the load is removed. How does the final length of the bar compare with its original length?



【中文翻譯】

一桿長0.8 m，由一結構鋼製成，具如圖示之應變 - 應力曲線。該鋼的降伏應力為250 MPa，且彈性模為200 GPa。此桿受軸向負載，一直到伸長為2.5 mm，然後移除負載。最後的尺寸與原本的尺寸相比較，結果如何？

【解析】



$$\varepsilon_B = \frac{\delta}{L} = \frac{2.5}{800} = 0.003125 \Rightarrow \text{塑性變形}$$

$$\varepsilon_E = \frac{\sigma_B}{E} = \frac{250(MPa)}{200(GPa)} = 0.00125$$

$$\varepsilon_r = \varepsilon_B - \varepsilon_E = 0.003125 - 0.00125 = 0.001875$$

$$\delta = L\varepsilon_r = 800 \times 0.001875\,(mm) = 1.5(mm)$$

$$L = L' - \delta = 250(mm) - 1.825 = 254.175(mm)$$

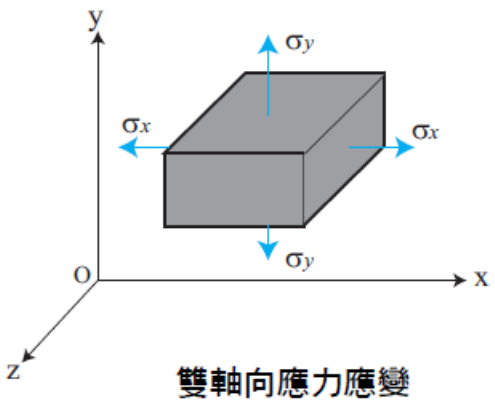
4-3. 應力與應變轉換

一、 應力與應變轉換

當一物體同時承受數個負荷作用時，其數個負荷所產生的效應會等於各個負荷單獨作用所生的效應之和，此原理稱之爲重疊原理(method of superposition)。

1.二軸向平面應力應變

如圖　所示，在二軸向平面應力應變中，當物體受到相互正交的二軸應力σ_x與σ_y作用時，會同時在X與Y方向產生相對應的應變ε_x與ε_y，在X方向的應力會在X方向產生一伸長的應變，但在Y方向則會產生一縮短的應變，爲說明方便，對於連續均質各向同性線彈性材料，可以將這種應力狀態，視爲二個單向應力狀態疊加來求主應變，在σ_x單獨作用下，沿主應力σ_x、σ_y方向的線應變分別爲：



雙軸向應力應變

$$\varepsilon'_x = \frac{\sigma_x}{E}\ ,\ \varepsilon'_y = -\frac{\nu\sigma_x}{E}$$

式中E、ν爲材料的彈性模數及蒲松比（Poisson ratio）。

同理，在σ_y單獨作用時，上述應變分別爲：

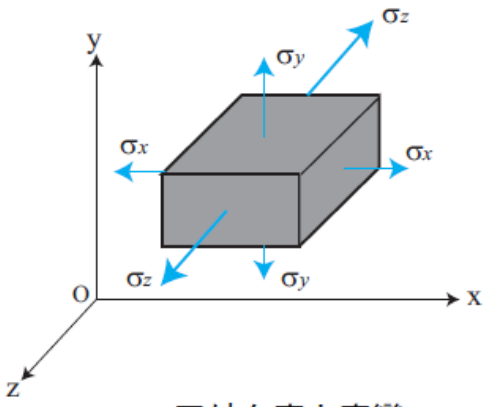
$$\varepsilon''_x = -\frac{\nu\sigma_x}{E}\ ,\ \varepsilon''_y = \frac{\sigma_y}{E}$$

故 $\varepsilon_x = \varepsilon'_x + \varepsilon''_x = \frac{\sigma_x}{E} - \frac{\nu\sigma_y}{E}$ ； $\varepsilon_y = \varepsilon'_y + \varepsilon''_y = \frac{\sigma_y}{E} - \frac{\nu\sigma_x}{E}$

二、 廣義的虎克定律

2.三軸向平面應力應變

在三向應力狀態下主單元體同時受到主應力σ_x、σ_y及σ_z作用，如圖1.4所示，我們把沿單元體主應力方向的線應變稱爲主應變（principal strain），習慣上分別用ε_x、ε_y及ε_z來表示，對於連續均質各向同性線彈性材料，可以將這種應力狀態，視爲三個單向應力狀態疊加來求主應變，將同方向的線應變疊加得三向應力狀態下主單元體的主應變爲：



三軸向應力應變

$$\begin{cases} \varepsilon_x = \dfrac{1}{E}\left[\sigma_x - \nu(\sigma_y + \sigma_z)\right] \\ \varepsilon_y = \dfrac{1}{E}\left[\sigma_y - \nu(\sigma_x + \sigma_z)\right] \\ \varepsilon_z = \dfrac{1}{E}\left[\sigma_z - \nu(\sigma_x + \sigma_y)\right] \end{cases} \qquad \begin{cases} \sigma_x = \dfrac{E}{(1+\nu)(1\text{-}2\nu)}\left[(1-\nu)\varepsilon_x + \nu(\varepsilon_y + \varepsilon_z)\right] \\ \sigma_y = \dfrac{E}{(1+\nu)(1-2\nu)}\left[(1-\nu)\varepsilon_y + \nu(\varepsilon_x + \varepsilon_z)\right] \\ \sigma_z = \dfrac{E}{(1+\nu)(1-2\nu)}\left[(1-\nu)\varepsilon_z + \nu(\varepsilon_x + \varepsilon_y)\right] \end{cases}$$

【觀念說明】

一根任意材質的棒子受均勻軸向應力試計算波松比的最大值。

利用一般虎克定律

$$\varepsilon_x + \varepsilon_y + \varepsilon_z = \frac{1-2\nu}{E}\left(\sigma_x + \sigma_y + \sigma_z\right) \qquad \text{(a)}$$

對常數三軸應力來說：

$$\varepsilon_x = \varepsilon_y = \varepsilon_z = \varepsilon, \quad \sigma_x = \sigma_y = \sigma_z = \sigma$$

所以，式(a)變成 $\varepsilon = \dfrac{1-2\nu}{E}\sigma$.由於 σ 及 ε 必為相同正負值：

$1 - 2\nu \geq 0$ or $\nu = \frac{1}{2}$

3.**體積應變及體積彈性係數**

材料的單位體積爲$V = dx \cdot dy \cdot dz$，若該單位體積受力變形後，其體積爲

$$V_1 = (dx + \varepsilon_x dx) \cdot (dy + \varepsilon_y dy) \cdot (dz + \varepsilon_z dz)$$

體積應變爲

$$\varepsilon_v = \frac{V_1 - V}{V} = (1+\varepsilon_x)(1+\varepsilon_y)(1+\varepsilon_z) - 1 \approx \varepsilon_x + \varepsilon_y + \varepsilon_z$$

$$\Rightarrow \frac{1-2v}{E}(\sigma_x + \sigma_y + \sigma_z) = \frac{3(1-2v)}{E}\left(\frac{\sigma_x + \sigma_y + \sigma_z}{3}\right) = \frac{\sigma_m}{K}$$

其中$K = \dfrac{E}{3(1-2v)}$爲體積彈性模數，$\sigma_m = \dfrac{\sigma_x + \sigma_y + \sigma_z}{3}$是三個主應力的平均值。

4.**溫度效應**

(1)總應變＝應力應變＋熱應變

$$\begin{cases} \varepsilon_x = \dfrac{1}{E}\left[\sigma_x - \nu(\sigma_y + \sigma_z)\right] + \alpha\,\Delta T \\ \varepsilon_y = \dfrac{1}{E}\left[\sigma_y - \nu(\sigma_x + \sigma_z)\right] + \alpha\,\Delta T \\ \varepsilon_z = \dfrac{1}{E}\left[\sigma_z - \nu(\sigma_x + \sigma_y)\right] + \alpha\,\Delta T \end{cases}$$

其中α爲熱膨脹係數、ΔT爲升高之溫度

(2)應變能密度＝$\dfrac{1}{2}$應力×應力應變＝$\dfrac{1}{2}\sigma(\varepsilon_x - \alpha\,\Delta T)$

三、剪應力與剪應變的虎克定律

剪應力與剪應變的虎克定律

1. 在剪應力不超過材料的降伏剪應力τ_y時，剪應力與剪應變成正比，即當$\tau \le \tau_y$時，$\tau = G\gamma$。
2. G—剪力彈性模數，對各向同性材料，三個彈性常數 E、ν(蒲松比)、G 之間的關係爲：$G = \dfrac{E}{2(1+\nu)}$

證明:(105 台大土木、105 成大工科、105 成大土木)

如圖(a)所示考慮一平面純剪應力態，其中該剪應力為τ_0，其主應力態如圖(b)所示，其中材料點的主應力為σ_1及σ_2其值分別為

$\sigma_1 = \tau_0$、$\sigma_2 = -\tau_0$

(一)考慮(a)圖

$$\varepsilon_x = \frac{1}{E}[\sigma_x - \upsilon(\sigma_y)] = 0$$

$$\varepsilon_y = \frac{1}{E}[\sigma_y - \upsilon(\sigma_x)] = 0$$

由虎克定律

$$\gamma_{xy} = \frac{\tau_{xy}}{G} = \frac{\tau_0}{G}$$

$$\varepsilon_{45^\circ} = \frac{\gamma_{xy}}{2} \times \sin 90^\circ = \frac{\tau_0}{2G}$$

(二)考慮(b)圖

$$\varepsilon_1 = \frac{1}{E}[\sigma_1 - \upsilon(\sigma_2)] = \frac{\tau_0}{\mathrm{E}}(1+\upsilon)$$

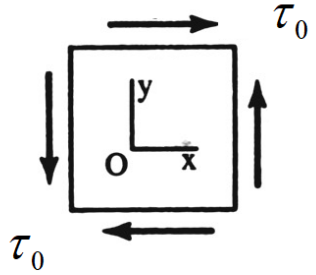
$$\varepsilon_2 = \frac{1}{E}[\sigma_2 - \upsilon(\sigma_1)] = \frac{-\tau_0}{\mathrm{E}}(1+\upsilon)$$

$$\gamma_{12} = 0$$

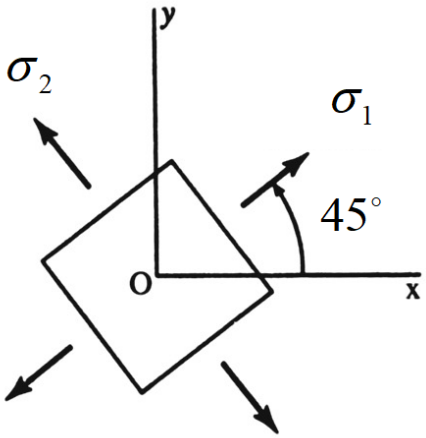
依應變轉換公式

$$\frac{\gamma_{xy}}{2} = -\frac{\varepsilon_1 - \varepsilon_2}{2} \times \sin(-90^\circ) + \frac{\gamma_{12}}{2}\cos(-90^\circ)$$

$$\frac{\tau_0}{G} = 2 \times \frac{\tau_0}{E}(1+\upsilon) \Rightarrow G = \frac{E}{2(1+\upsilon)}$$



(a)



(b)

四、應變能及應變能密度

(一)應變能密度(strain-energy density)

材料每單位體積之能量稱之為應變能密度 (strain-energy density)，而若材料承受一單軸向應力 σ ，則由式所定義之應變能可寫為

$$u = \int \sigma d\varepsilon \Rightarrow u = \frac{1}{2}\sigma\varepsilon = \frac{1}{2}E\varepsilon^2 = \frac{1}{2}\frac{\sigma^2}{E}$$

而對於三維問題應變能密度為

$$u = \frac{1}{2}\sigma_x\varepsilon_x + \frac{1}{2}\sigma_y\varepsilon_y + \frac{1}{2}\sigma_z\varepsilon_z + \frac{1}{2}\tau_{xy}\gamma_{xy} + \frac{1}{2}\tau_{xz}\gamma_{xz} + \frac{1}{2}\tau_{yz}\gamma_{yz}$$

其中分別為 σ 及 ε 正應力和正應變， τ 及 γ 分別為切應力和切應變。

(二)**應變能（strain energy）**

應變能或稱變形能，是指物體在發生形變時貯存於其中的能量，當外力撤去後，應變能會逐漸釋放，對於彈性材料，應變能可以完全恢復，物體將回到初始狀態，而對塑性材料而言，應變能無法完全恢復。應變能通常記作 U， 可表示為

U = 應力所做之功 = 內力所做之功 $\Rightarrow \mathrm{U} = \int u d\forall$

對於三維問題應變能為

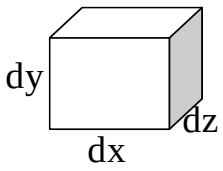
$$\mathrm{U} = \int \frac{1}{2}\left(\sigma_x\varepsilon_x + \sigma_y\varepsilon_y + \tau_{xy}\gamma_{yz} + \tau_{zx}\gamma_{zx}\right)\mathrm{d}\forall \text{其中} \mathrm{d}\forall = \mathrm{xdydz}$$

dx, dy, dz 為物體內一個小立方體之尺寸，dv 為此小立方的體積，

故 dv=dxdydz

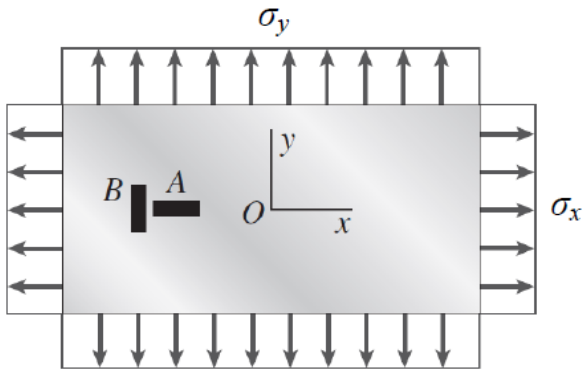
若以主應力來計算：

$$\mathrm{U} = \frac{1}{2\mathrm{E}}\int \sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu\left(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_1\sigma_3\right)\mathrm{dxdydz}$$



範例 4

A rectangular steel plate with thickness t=6.5mm. is subjected to uniform normal stresses x and y, as shown in the figure. Strain gages A and B, oriented in the x and y directions, respectively, are attached to the plate. The gage readings give normal strains $\varepsilon_x = 0.00062$ (elongation) and $\varepsilon_y = -0.00045$ (shortening). Knowing that $E = 210GPa$ and $\nu = 0.3$, determine the stresses σ_x and σ_y and the change Δt in the thickness of the plate.



【107 成大航太】【105 地特四等】【100 台科營建】【中興機械類題】

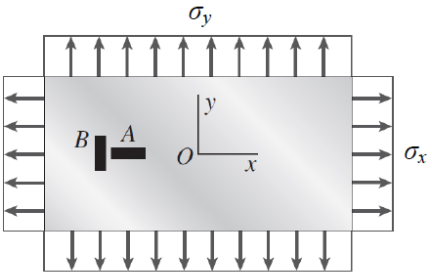
【中文翻譯】

如下圖所示，一長方形鋼板，厚度 $t = 6.5$ mm，受到均勻未知之正應力 σ_x 和 σ_y。兩應變規 A 和 B 分別沿著 x 方向和 y 方向於應力施加前貼於鋼板上，應力施加後，應變規 A 和 B 之應變讀數分別為 $\varepsilon_x = 0.00062$（伸長）和 $\varepsilon_y = -0.00045$（縮短）。已知鋼板之楊氏模數 $E = 210$ GPa，波松比（Poisson's ratio）$\nu = 0.3$。

㈠試求正應力 σ_x 和 σ_y。（10 分）

㈡試求鋼板厚度之變化量 Δt。（4 分）

㈢試求最大剪應力並繪製對應之應力元素圖。（6 分）



【解析】

已知應變求應力。

(一) $\sigma_x = \frac{E}{1-\upsilon^2}(\varepsilon_x + \nu\varepsilon_y) = 111.9(\text{Mpa})$

$\sigma_y = \frac{E}{1-\upsilon^2}(\nu\varepsilon_x + \varepsilon_y) = -60.9(\text{Mpa})$

(二) $\varepsilon_z = \frac{-\upsilon}{E}(\sigma_x + \sigma_y) = -72.86\mu$

$\Delta z = \varepsilon_z \times t = -473.57\mu(\text{mm})$ 負號表示矩形鋼板厚度變薄

(三) $\tau_{max} = \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + \tau_{xy}^2} = 86.4(\text{Mpa})$

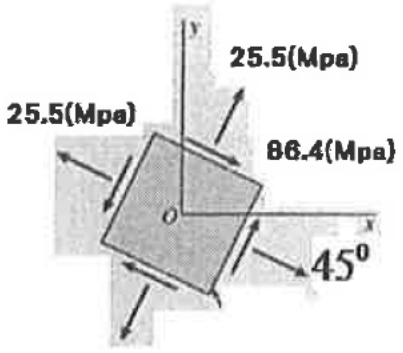
$\sigma_s = \frac{\sigma_x + \sigma_y}{2} = 25.5(\text{Mpa})$

由式 $\tau_\theta = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$

知當 $\sin 2\theta = -1$ 時 τ_θ 存在最大值此時

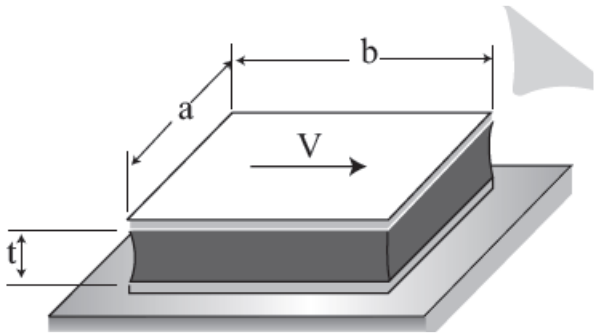
$\theta = -45°$ (順轉45度)

$\tau_{\theta=45} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta = 86.4(\text{Mpa})$



範例 5

有一長方體，長、寬、高分別為b=250mm、a=125mm、t=50mm，底部面固定於地上且頂部面移動8mm，如圖所示，今以12KN之V力推之，試求剪力彈性係數G 為多少？

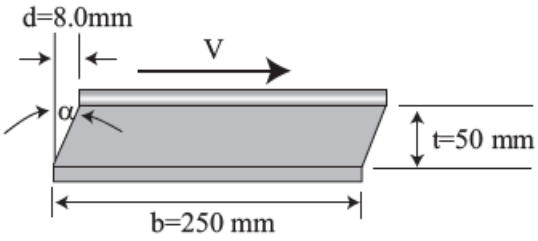


【解析】

$$\gamma_{ave}=\frac{d}{t} \qquad \gamma_{ave}=0.16$$

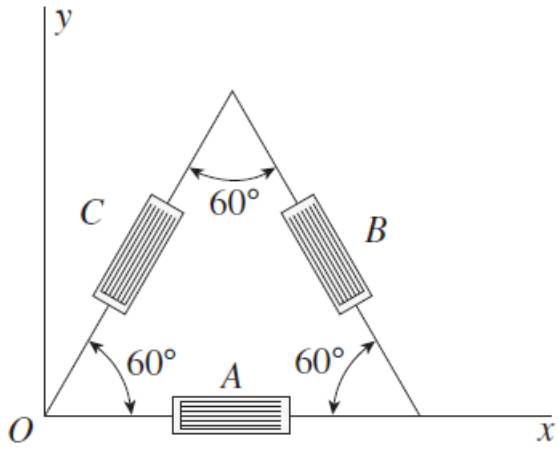
$$G=\frac{\tau_{ave}}{\gamma_{ave}}=\frac{0.384}{0.16}=2.4MPa$$

$$T_{av}=\frac{12k}{125\times 250}=0.384(MPa)$$



範例 6

The strain rosette shown was used to obtain the following strain data at a point on the free surface of an aluminum alloy ($E = 70$ GPa, and $\nu = 0.33$) machine part: $\varepsilon_a = +2000\mu$, $\varepsilon_b = +1500\mu$, $\varepsilon_c = -1300\mu$ Determine the principal stresses and the maximum shearing stress at the point by first computing the principal strains and the maximum shearing strain at the point and using these strains to compute the stresses.



【99 清大動機】

【中文翻譯】

一鋁合金(E = 70GPa，ν = 0.33)構件之表面用圖中60°菊花型應變規測得之正交應變為ε_a = 2000μ，ε_b = 1500μ，ε_c = −1300μ，試求在此點之主應力、最大剪應力、主應變、最大剪應變。

【解析】

(一)

$$\varepsilon_x = 2000\,\mu = \varepsilon_a$$

$$\varepsilon_{120^\circ} = \varepsilon_b = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = 1500\mu$$

$$\Rightarrow \frac{2000 + \varepsilon_y}{2} + \frac{2000 - \varepsilon_y}{2}\cos 2\times(120^\circ) + \frac{\gamma_{xy}}{2}\sin 2\times(120^\circ) = 1500\mu$$

同理

$$\varepsilon_{60^\circ} = \varepsilon_c = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = -1300\mu$$

$$\Rightarrow \frac{2000 + \varepsilon_y}{2} + \frac{2000 - \varepsilon_y}{2}\cos 2\times(60^\circ) + \frac{\gamma_{xy}}{2}\sin 2\times(60^\circ) = -1300\mu$$

由上兩式可解得

$\varepsilon_y = -533.3\mu$ 、 $\gamma_{xy} = -3233\,\mu$

(二)

$$\frac{\gamma_{\max}}{2} = \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2} \Rightarrow \gamma_{\max} = 4107\mu$$

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2} = 2787\mu, -1320\mu$$

由虎克定律

$$\varepsilon_1 = \frac{1}{E}(\sigma_1 - \upsilon\sigma_2) \Rightarrow 2787\mu = \frac{1}{70\times10^{-3}}(\sigma_1 - 0.33\sigma_2)$$

$$\varepsilon_2 = \frac{1}{E}(\sigma_2 - \upsilon\sigma_1) \Rightarrow -1320\mu = \frac{1}{70\times10^{-3}}(\sigma_2 - 0.33\sigma_1)$$

$$\sigma_1 = 184.7(Mpa), \sigma_2 = -31.4(Mpa)$$

$$G = \frac{E}{2(1+\upsilon)} = 26.32(GPa)$$

$$\tau_{\max} = G\gamma_{\max} = 108.1(GPa)$$

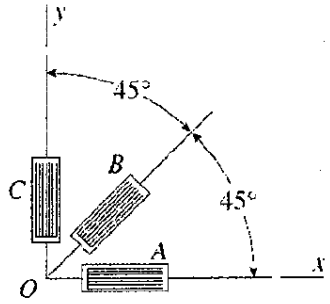
範例 7

(20%) During a test of an airplane wing, the strain gage readings from a 45° rosette (see figure) are as follows: gage A, 500×10^{-6}; gage B, 350×10^{-6}; and gage C, -100×10^{-6}. The wing is made of an aluminum alloy having $E=100\text{GPa}$ and $\nu=0.3$.

(a) Determine the principal strains and maximum shear strain.

(b) Determine the principal stresses and maximum shear stress.

(c) Discuss whether the orientation of principal strains and principal stresses are the same.



【105 成大航太】

【解析】

$\varepsilon_y=-100\mu=\varepsilon_c$ 、 $\varepsilon_x=500\mu=\varepsilon_a$

$$\varepsilon_\theta=\varepsilon_b=\frac{\varepsilon_x+\varepsilon_y}{2}+\frac{\varepsilon_x-\varepsilon_y}{2}\cos2\theta+\frac{\gamma_{xy}}{2}\sin2\theta=350\mu$$

$$\Rightarrow\frac{500-100}{2}+\frac{500-(-100)}{2}\cos(2\times45^\circ)+\frac{\gamma_{xy}}{2}\sin(2\times45^\circ)=350\mu\Rightarrow\gamma_{xy}=300\mu$$

主應變 $\varepsilon_{1,2}=\dfrac{\varepsilon_x+\varepsilon_y}{2}\pm\sqrt{(\dfrac{\varepsilon_x-\varepsilon_y}{2})^2+(\dfrac{\gamma_{xy}}{2})^2}\Rightarrow\varepsilon_1=535.41\mu,\varepsilon_2=-135.41\mu$

$$\frac{\gamma_{\max}}{2}=\frac{\varepsilon_1-\varepsilon_2}{2}\Rightarrow\gamma_{\max}=670.82\mu$$

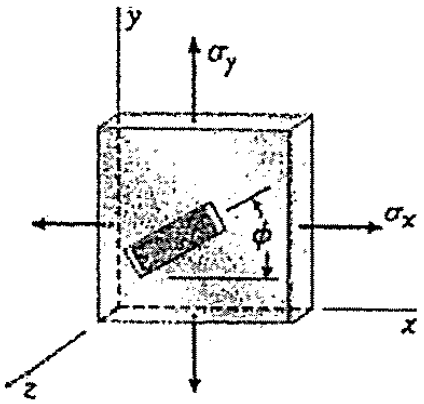
$$\sigma_1=\frac{E}{1-\upsilon^2}(\varepsilon_1+\upsilon\varepsilon_2)=54.37(MPa)$$

同理

$\sigma_2=2.77(MPa)$ 、 $\tau_{\max}=\dfrac{\sigma_1-\sigma_2}{2}=25.8(MPa)$

範例 8

A steel plate with modulus of elasticity $E = 200\,\text{GPa}$ and Poisson's ratio $\nu = 0.30$ is loaded in by normal stresses σ_x and σ_y (see figure). A strain gage is bonded to the plate at an angle $\phi = 30^\circ$ .If the stress σ_x is 125 MPa and the strain measured by the gage is $\varepsilon = 407 \times 10^{-6}$,what is the maximum in-plane shear stress $(\tau_{\max})_{xy}$ and shear strain $(\gamma_{\max})_{xy}$? What is themaximum shear strain $(\gamma_{\max})_{xz}$ in the *xz* plane? What is the maximum shear strain $(\gamma_{\max})_{yz}$ in the *yz* plane?



【105 台大土木大地組】

【中文翻譯】

具彈性模數 $E = 200\,\text{GPa}$ 及蒲松比 $\nu = 0.30$ 的鋼板，承受雙軸向應力以正向應力 σ_x 及 σ_y （ 見圖 ）。應變規以角度 $\phi = 30^\circ$ 貼在板上。如果應力 σ_x 為125 MPa，具規所量得之應變為 $\varepsilon = 407 \times 10^{-6}$ ，最大平面上剪應力 $(\tau_{\max})_{xy}$ 及剪應變 $(\gamma_{\max})_{xy}$ 為何？ *xz* 平面之最大剪應變 $(\gamma_{\max})_{xz}$ 為何？ *yz* 平面之最大剪應數 $(\gamma_{\max})_{yz}$ 為何？

【解析】

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y) = \frac{1}{200000}(125 - 0.3\sigma_y)$$

$$\varepsilon_y = \frac{1}{E}(\sigma_y - \nu\sigma_x) = \frac{1}{200000}(\sigma_y - 37.5)$$

$$\varepsilon_z = -\frac{\nu}{E}(\sigma_x + \sigma_y) = -\frac{0.3}{200000}(125 + 0.3\sigma_y)$$

$$\varepsilon_{x_1} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

$$407 \times 10^{-6} = \left(\frac{1}{2}\right)\left(\frac{1}{200000}\right)(87.5 + 0.7\sigma_y) + \left(\frac{1}{2}\right)\left(\frac{1}{200000}\right)(162.5 - 1.3\sigma_y)\cos 60^\circ$$

$$\sigma_y = -119 \text{ MPa}$$

$$(\tau_{max})_{xy} = \frac{\sigma_x - \sigma_y}{2} = 122 \text{ MPa}$$

$$\varepsilon_x = 803.5 \times 10^{-6},\ \varepsilon_y = -782.5 \times 10^{-6},\ \varepsilon_z = -9 \times 10^{-6}$$

xy plane: $\dfrac{(\gamma_{max})_{xy}}{2} = \sqrt{\left(\dfrac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\dfrac{\gamma_{xy}}{2}\right)^2}$

$\gamma_{xy} = 0,\ (\gamma_{max})_{xy} = 1586 \times 10^{-6}$

xz plane: $\dfrac{(\gamma_{max})_{xz}}{2} = \sqrt{\left(\dfrac{\varepsilon_x - \varepsilon_z}{2}\right)^2 + \left(\dfrac{\gamma_{xz}}{2}\right)^2}$

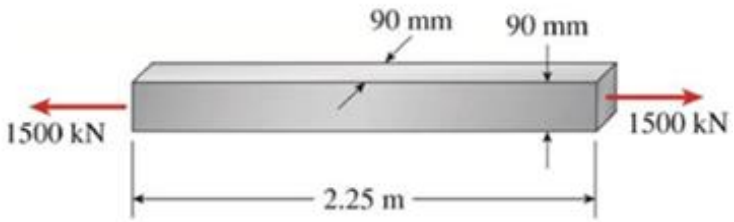
$\gamma_{xz} = 0,\ (\gamma_{max})_{xz} = 812.5 \times 10^{-6}$

yz plane: $\dfrac{(\gamma_{max})_{yz}}{2} = \sqrt{\left(\dfrac{\varepsilon_y - \varepsilon_z}{2}\right)^2 + \left(\dfrac{\gamma_{yz}}{2}\right)^2}$

$\gamma_{yz} = 0,\ (\gamma_{max})_{yz} = 773.5 \times 10^{-6}$

範例 9

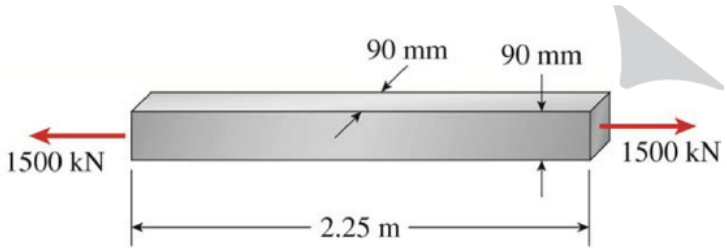
A bar of length 2.25 m with a square cross section 90 mm on each side is subjected to an axial tensile force of 1500 kN (see figure). Assume that E= 110 GPa and v=0.34. Determine the increase in volume of the bar



【104 地特四等】【103 成大醫工】

【中文翻譯】

如圖所示之黃銅棒具有90mm邊長之正方形截面，長度為2.25m，承受1500kN之軸向負載，若材料之楊氏係數E=110GPa、波松比v=0.34，試求黃銅棒受力所產生之體積變化量。



【解析】

$$\varepsilon_{\forall}=\frac{1-2V}{E}(\sigma_x+\sigma_y+\sigma_z)=\frac{1-2(0.34)}{110\times10^3}\times(\frac{1500\times10^3}{90\times90})$$

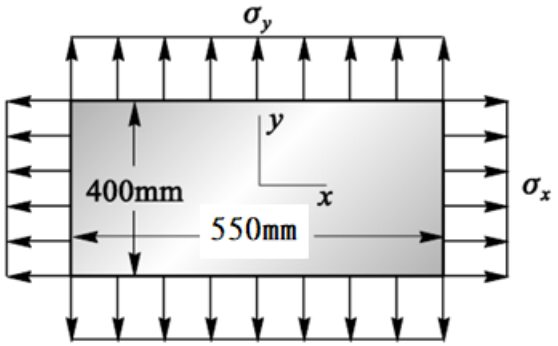
$$=5.3872\times10^{-4}$$

$$\varepsilon_{\forall}=\frac{\forall'}{\forall}=5.3872\times10^{-4}\Rightarrow\forall'=5.3872\times10^{-4}\times90\times90\times2250$$

$$=9818.18(mm^3)$$

範例 10

A rectangular plate in biaxial stress (see figure) is subjected to normal stresses $\sigma_x = 67(MPa)$ (tension) and $\sigma_y = -23(MPa)$ (compression). The plate has dimensions $400 \times 550 \times 20mm$ and is made of steel with E=200GPa and $\upsilon = 0.3$. (a) Determine the maximum in-plane shear strain in the plate. (b) Determine the change in the thickness of the plate. (c) Determine the change in the volume of the plate



【106 台大工科】

【中文翻譯】

受雙軸向應力矩形板受有正向應力 $\sigma_x = 67(MPa)$ MPa（拉伸）及 $\sigma_y = -23(MPa)$（壓縮）。板之尺寸為 $400 \times 550 \times 20mm$，由鋁所製成，具 E=200GPa 及 $\upsilon = 0.3$。(a) 求板之最大平面上剪應力 γ_{max}。(b) 求板厚度之變化 Δt。(c) 求板體積之改變 ΔV。

【解析】

$\sigma_x = \sigma_1 = 67(MPa)$ MPa（拉伸）及 $\sigma_y = \sigma_2 = -23(MPa)$（壓縮）

E=200GPa及 $\upsilon = 0.3$

$$G = \frac{E}{2 \times (1 + \upsilon)} = 76.92(MPa)$$

(a)

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2} = 45(MPa)$$

$$\tau_{max} = G\gamma_{max} \Rightarrow \gamma_{max} = 585 \times 10^{-6}$$

(b)

$$\varepsilon_z = \frac{1}{E}[\sigma_z - \upsilon(\sigma_x + \sigma_y)] \Rightarrow \varepsilon_z = \frac{1}{200 \times 10^3}[0 - 0.3(67 + (-23)] = 6.6 \times 10^{-5}$$

$$\Delta t = \varepsilon_z t = 1.32 \times 10^{-3} mm$$

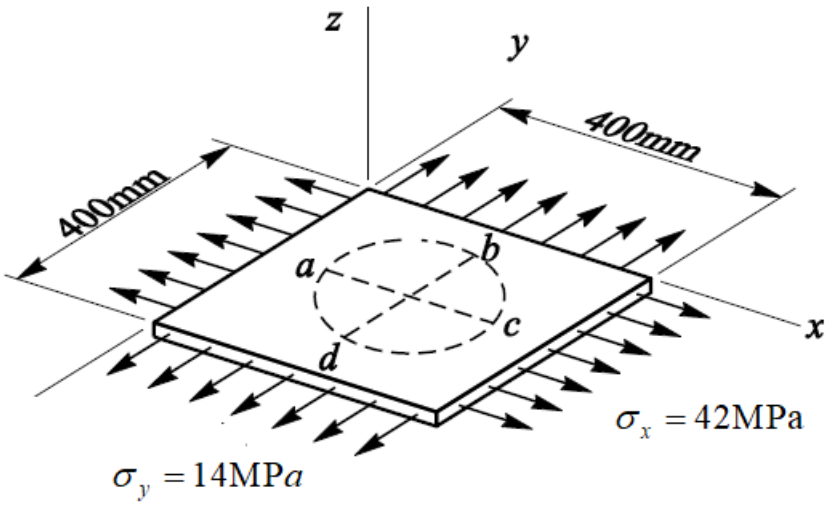
(c)

$$V_0 = 400 \times 550 \times 20mm = 4.4 \times 10^6 (mm^3)$$

$$\Delta V = V_0 (\frac{1-2\upsilon}{E})(\sigma_x + \sigma_y) = 4.4 \times 10^6 \times (\frac{1-2 \times 0.3}{200000})(67-23) = 387.2(mm^3)$$

範例 11

A circle of diameter $d = 200\ \text{mm}$ is etched on a brass plate (see figure). The plate has dimensions $400 \times 400 \times 20\text{mm}$. Forces are applied to the plate, producing uniformly distributed normal stresses $\sigma_x = 42\text{MPa}$ and $\sigma_y = 14\text{MP}a$. Calculate the following quantities: (a) the change in length Δac of diameter ac; (b) the change in length Δbd of diameter bd; (c) the change Δt in the thickness of the plate; (d) the change ΔV in the volume of the plate, and (e) the strain energy U stored in the plate. (Assume E =100 GPa and $v = 0.34$)



【106 北科】【106 中央土木】

【中文翻譯】

圓直徑 $d = 200\ \text{mm}$，蝕刻在黃銅板上（見圖）。板的尺寸為 $400 \times 400 \times 20\ \text{mm}$，力施於板，產生均勻分佈之正向應力 $\sigma_x = 42\text{MPa}$ 及 $\sigma_y = 14\text{MP}a$ MPa。求如下量：(a) 直徑 ac 之長度的改變 Δac；(b) 直徑 bd 的長度之改變 Δbd；(c) 板厚的變化 Δt；(d) 板的體積的改變 ΔV；及 (e) 貯於板中的應變能 U。（假設 $E = 100\ \text{GPa}$ 及 $v = 0.34$）。

【解析】

(a)由虎克定律

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y) = 372.4 \times 10^{-6}$$

$$\Delta ac = \varepsilon_x d = 0.0745\ \text{mm}$$

(b)

$$\varepsilon_y = \frac{1}{E}(\sigma_y - \nu\sigma_x) = -2.80 \times 10^{-6}$$

$$\Delta bd = \varepsilon_v d = -560 \times 10^{-6} \text{ mm}$$

$$\varepsilon_z = -\frac{\nu}{E}(\sigma_x + \sigma_y) = -190.4 \times 10^{-6}$$

$$\Delta t = \varepsilon_z t = -0.00381 \text{ mm}$$

(d)

$$e = \frac{1 - 2\nu}{E}(\sigma_x + \sigma_y) = 179.2 \times 10^{-6}$$

$$V_0 = (400)(400)(20) = 3.2 \times 10^6 \text{ mm}^3$$

$$\Delta V = eV_0 = 573 \text{ mm}^3$$

(e)

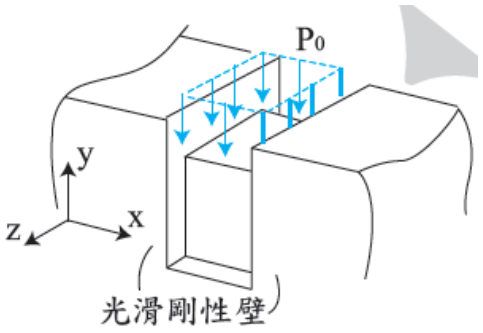
$$u = \frac{1}{2E}(\sigma_x^2 + \sigma_y^2 - 2\nu\sigma_x\sigma_y)$$

$$= 7.801 \times 10^{-3} \text{ MPa}$$

$$U = uV_0 = 25.0 \text{ N} \cdot \text{m} = 25.0 \text{ J}$$

範例 12

如圖所示，物體置於兩平行的光滑剛性壁之間，材料在z方向不受拘束，若頂部施加壓力P_0且材料彈性模數為E、蒲松比 ν，試以P_0、E、ν求出 (1)此物體與光滑剛性壁之間的側向壓力。(2)若應變很小求出單位體積。



【解析】

(1) 如圖所示

在z方向之應力 $=0$且x方向之應變 $=0$

$$\Rightarrow \sigma_z = 0\ ,\ \varepsilon_x = 0\ ,\ \sigma_y = -P_o \qquad \varepsilon_x = \frac{1}{E}[\sigma_x - \nu(\sigma_y + \sigma_z)] = 0$$

$$\Rightarrow \sigma_x = \nu\,\sigma_y = -P_o\nu$$

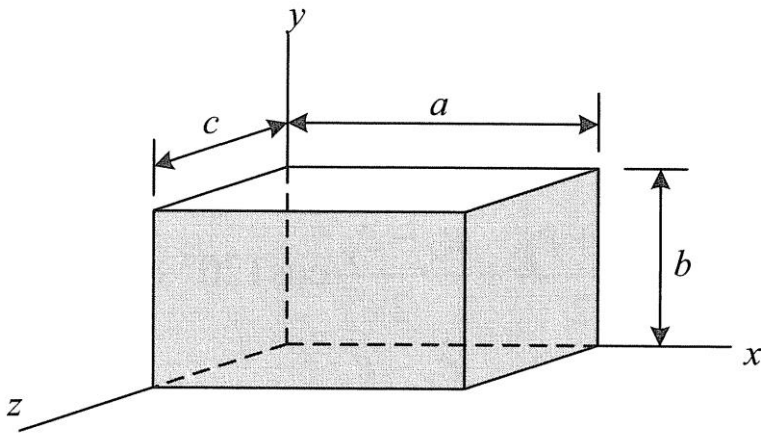
(2) $c = \varepsilon_x + \varepsilon_y + \varepsilon_z$

$$= \frac{\sigma_y}{E} - \frac{\nu}{E}(\sigma_x + \sigma_z) + \frac{\sigma_z}{E} - \frac{\nu}{E}(\sigma_x + \sigma_y) = \frac{1}{E}(-P_o + \nu^2 P_o + \nu^2 P_o + \nu P_o)$$

$$= \frac{-P_o}{E}(1+\nu)(1-2\nu)$$

範例 13

圖中長方體鋼塊(E =200 GPa，ν = 0.30)之尺寸 a = 250 mm，b = 150 mm，c = 150 mm。鋼塊在 x、y 與 z 面上承受之正交應力分別為σ_x = −50 MPa、σ_y = −40 MPa 與σ_z = −36 MPa，試求(a)最大剪應力τ_{max}，(b)邊長之變化量Δa、Δb 與Δc，(c)體積之變化量ΔV(d)儲存之應變能



【成大航太】【交大機械】

【解析】

$\sigma_x = -50\,\text{MPa}$，$\sigma_y = -40\,\text{MPa}$，$\sigma_z = -36\,\text{MPa}$

(a)主應力：$\sigma_1 = -36\,\text{MPa}$，$\sigma_2 = -40\,\text{MPa}$，$\sigma_3 = -50\,\text{MPa}$

$$\tau_{\max} = \frac{1}{2}(\sigma_1 - \sigma_3) = \frac{1}{2}[(-36) - (-50)] = 7.0\ \text{MPa}$$

(b) $$\varepsilon_x = \frac{\sigma_y - \nu(\sigma_y + \sigma_z)}{E} = \frac{(-50) - 0.30[(-40) + (-36)]}{200\times10^3} = -0.136\times10^{-3}$$

$$\varepsilon_y = \frac{\sigma_y - \nu(\sigma_x + \sigma_z)}{E} = \frac{(-40) - 0.30[(-50) + (-36)]}{200\times10^3} = -0.0710\times10^{-3}$$

$$\varepsilon_z = \frac{\sigma_z - \nu(\sigma_x + \sigma_y)}{E} = \frac{(-36) - 0.30[(-50) + (-40)]}{200\times10^3} = -0.0450\times10^{-3}$$

$$\Delta a = \varepsilon_x a = (-0.136\times10^{-3})(250) = -0.0340\,\text{mm}$$

$$\Delta b = \varepsilon_y b = (-0.0710\times10^{-3})(150) = -0.01065\,\text{mm}$$

$$\Delta c = \varepsilon_z c = (-0.0450\times10^{-3})(150) = -0.00675\,\text{mm}$$

(c) $$e = \varepsilon_x + \varepsilon_y + \varepsilon_z = (-0.136 - 0.0710 - 0.0450)\times10^{-3} = -0.252\times10^{-3}$$

$$\Delta V = eV = (-0.252\times10^{-3})(250\times150\times150) = -1418\,\text{mm}^3$$

(d) $$u = \frac{1}{2}(\sigma_x\varepsilon_x + \sigma_y\varepsilon_y + \sigma_z\varepsilon_z)$$

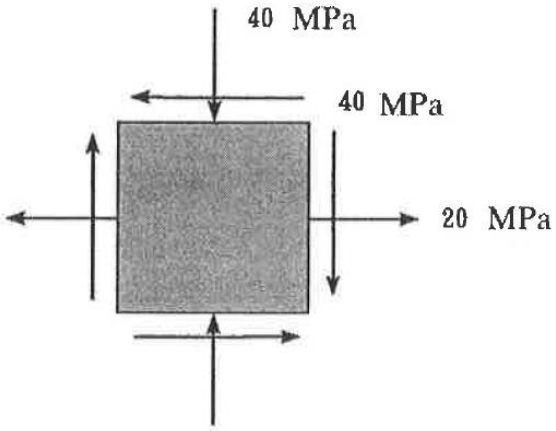
$$= \frac{1}{2}[(-50)(-0.136) + (-40)(-0.0710) + (-36)(-0.0450)]\times10^{-3} = 5.63\times10^{-3}\ \text{N/mm}^2$$

$$U = uV = (5.63\times10^{-3})\,(250 \times 150 \times 150) = 31.7 \times 10^3\ \text{N-mm} = 31.7\ \text{J}$$

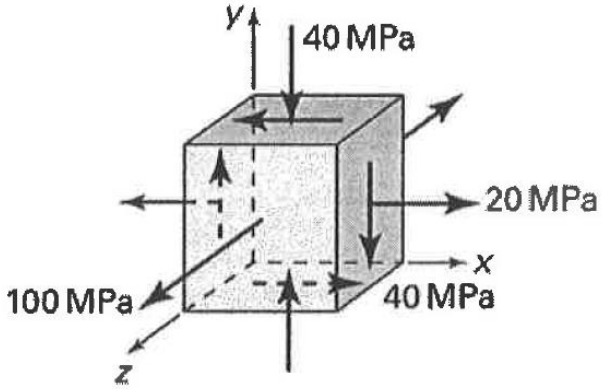
範例 14

圖三(a)所示為 x-y 平面應力問題的應力狀態，圖三(b)所示為將(a)的應力狀態額外在 z 軸方向再加上 100 MPa 拉應力後之三軸應力狀態。請計算下列性質；

1. x-y 平面應力情形下(圖 a)，主軸正應力大小及方向。(10%)
2. x-y 平面應力情形下(圖 a)，主軸剪應力大小。(5%)
3. 三軸應力情形下(圖 b)，絕對最大剪應力大小(absolute maximum shear stress)。(5%)
4. 假設彈性行為，已知材料性質 E=200 GPa, ν=0.3。三軸應力情形下(圖 b)，單位體積應變 *e* (volume change or *dilatation*)是多少? (5%)



(a) x-y 平面應力狀態



(b) 三軸應力狀態

圖三

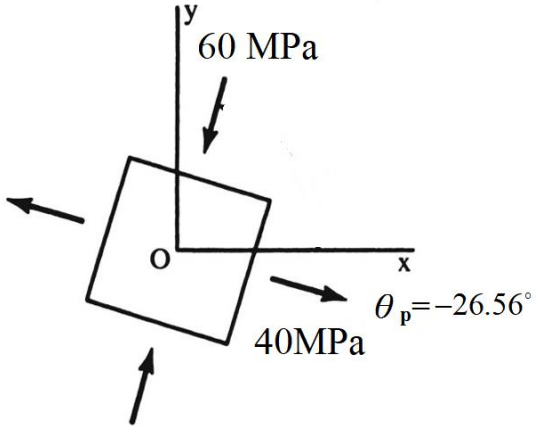
【105 中興土木】

【解析】

(1)x-y 平面主應力及最大剪應力

$$\sigma_{1,2}=\frac{\sigma_x+\sigma_y}{2}\pm\sqrt{(\frac{\sigma_x-\sigma_y}{2})^2+(\tau_{xy})^2}=\frac{20+(-40)}{2}\pm\sqrt{(\frac{20-(-40)}{2})^2+(-40)^2}=40,-60(MPa)$$

$$\tan(2\theta_p)=\frac{2\tau_{xy}}{\sigma_x-\sigma_y}\Rightarrow\ \theta_p=\frac{1}{2}\tan^{-1}(\frac{2\tau_{xy}}{\sigma_x-\sigma_y})=-26.56^\circ$$



(2)

主軸剪應力為零

(3)

$\sigma_{1,2} = 40,-60(MPa)$ 、 $\sigma_3 = 100(MPa)$

$$\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2} = \frac{100-(-60)}{2} = 80(MPa)$$

(4)

$$e = \frac{1-2\upsilon}{E}(\sigma_x + \sigma_y + \sigma_z) = \frac{1-2\times 0.3}{200\times 10^3}(20+(-40)+100) = 1.6\times 10^{-4}$$

範例 15

假設有一體積為$1cm^3$之黃銅置於剛性鉗夾之間並加熱 100℃，不計摩擦，試決定 x、y、z 方向作用於自發體之應力，此黃銅 $E=105\times10^3(MPa)$、$\upsilon=0.33$、$\alpha=18.8\times10^{-6}/^\circ C$ (1)當鉗夾僅限制立方體 x 方向之運動(2) 鉗夾僅限制立方體 x 及 y 方向之運動(3) 鉗夾僅限制立方體 x、y、z 方向之運動

【大同機械】

【解析】

(1)

$\sigma_y=0$、$\sigma_z=0$

$$\varepsilon_x=\frac{1}{E}\times[\sigma_x]+\alpha\Delta T=0$$

$$\sigma_x=-E\alpha\Delta T=-105\times10^3\times100\times18.8\times10^{-6}=-197.4(MPa)$$

(2)

$\sigma_y=\sigma_x$、$\sigma_z=0$

$$\varepsilon_x=\frac{1}{E}\times[\sigma_x-\upsilon(\sigma_y)]+\alpha\Delta T=0$$

$$\sigma_x=\sigma_y=\frac{-E\alpha\Delta T}{1-\upsilon}=\frac{-105\times10^3\times100\times18.8\times10^{-6}}{1-0.33}=-294.6(MPa)$$

(3)

$\sigma_y=\sigma_x=\sigma_z$

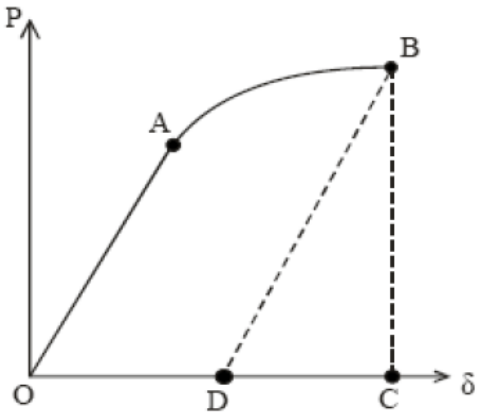
$$\varepsilon_x=\frac{1}{E}\times[\sigma_x-\upsilon(\sigma_y+\sigma_z)]+\alpha\Delta T=0$$

$$\sigma_x=\sigma_y=\frac{-E\alpha\Delta T}{1-2\upsilon}=\frac{-105\times10^3\times100\times18.8\times10^{-6}}{1-2\times0.33}=-580.6(MPa)$$

實戰演練

例 1

如圖所示，試以英文符號 OABCD 表示以下應變能面積：(一)彈性應變能(二)非彈性應變能(三)總應變能。(95 年機械地特四等)



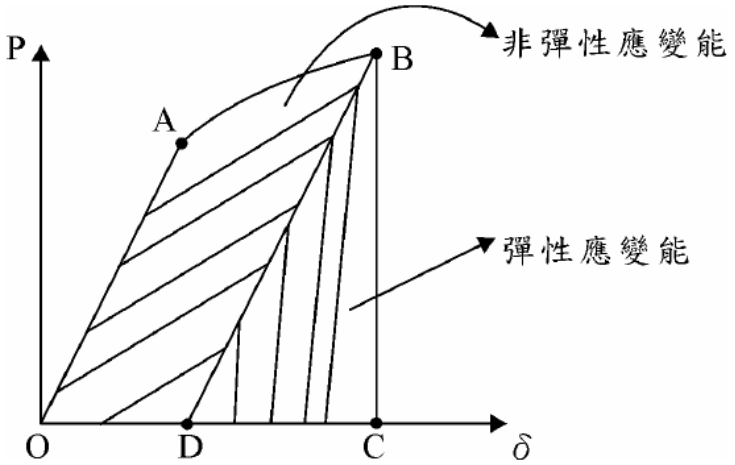
【解析】

1. (1) 當負載將桿拉伸時，即造成應變，這些應變使得本身的能量增加，稱之為應變能。

$$V = W = \int_o^{\delta} P d\delta$$

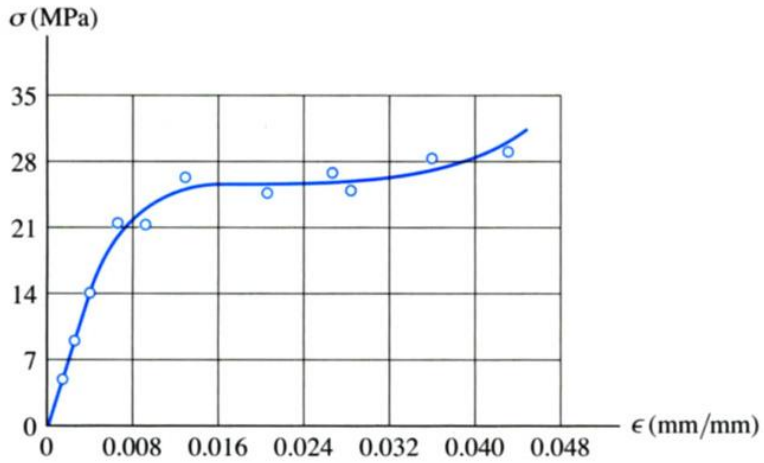
(2) 當加載時，負載所作的功等於曲線下的面積(面積 OABCDO)，稱之為總應變能

2. 當負載去除時，若 B 點超過彈性界限，則 P-δ 圖沿線 BD 並保持一永久伸長量 OD，因此卸載時所恢復的應變能(面積 DBCD)稱之為彈性應變能。
3. 面積 OABDO 代表桿的永久變形過程中所損失的能量，此能量稱之為非彈性應變能。



例 2

一用於同軸電纜外皮的聚乙烯之應力應變圖，是以一標距長度 250 mm 的試片做試驗所得到。若負載 P 導致試片產生應變 $\varepsilon = 0.024$ 對應 $\sigma = 25.55(MPa)$，當負載移除後，求兩標記點之間的近似長度。假設試片為彈性恢復。



【解析】

$$\mathrm{E} = \frac{\sigma}{\varepsilon} = \frac{14}{0.004} = 350(MPa)$$

$$\varepsilon = 0.024 \Rightarrow \sigma = 25.55(MPa)$$

$$L' = 250(mm) + 0.024 \times 250 = 256(mm)$$

$$\varepsilon_r = \frac{\sigma}{E} = \frac{25.55(MPa)}{3500(MPa)} = 0.0073$$

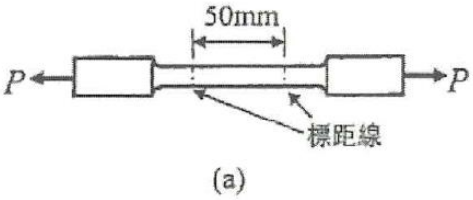
$$\delta = L\varepsilon_r = 1.825(mm)$$

$$L = L' - \delta = 250(mm) - 1.825 = 254.175(mm)$$

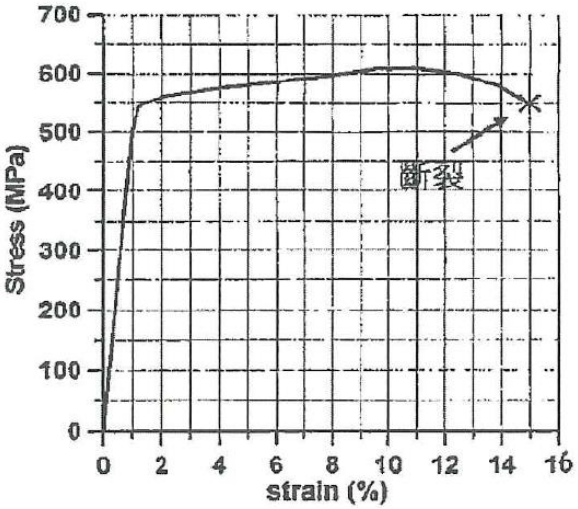
例 3

下圖(a)為一圓桿試件進行拉伸實驗的示意圖，假設在進行實驗前先在試桿上畫下距離為50mm 的標距線如圖所示，標距範圍之直徑為10mm，測試所得的拉伸曲線如圖(b)，設測試時試桿於標距範圍內斷裂，測試後將斷裂的兩半試桿緊密拼合在一起但不施加外力，則標距線的距離將為：

(A) 800mm (B) 57.5mm (C) 56.95mm (D) 754mm (E)資料不足無法計算



(a)



【台大機械】

【解析】

$$\sigma = \mathrm{E}\varepsilon \Rightarrow \mathrm{E} = \frac{500}{0.01} = 50000(MPa)$$

$$\varepsilon_{\mathrm{P}} = \varepsilon - \varepsilon_y = 0.15 - \frac{550}{50000} = 0.139$$

$$L^{'} = L(1+\varepsilon_{\mathrm{P}}) = 50 \times (1+0.139) = 56.95(mm)$$

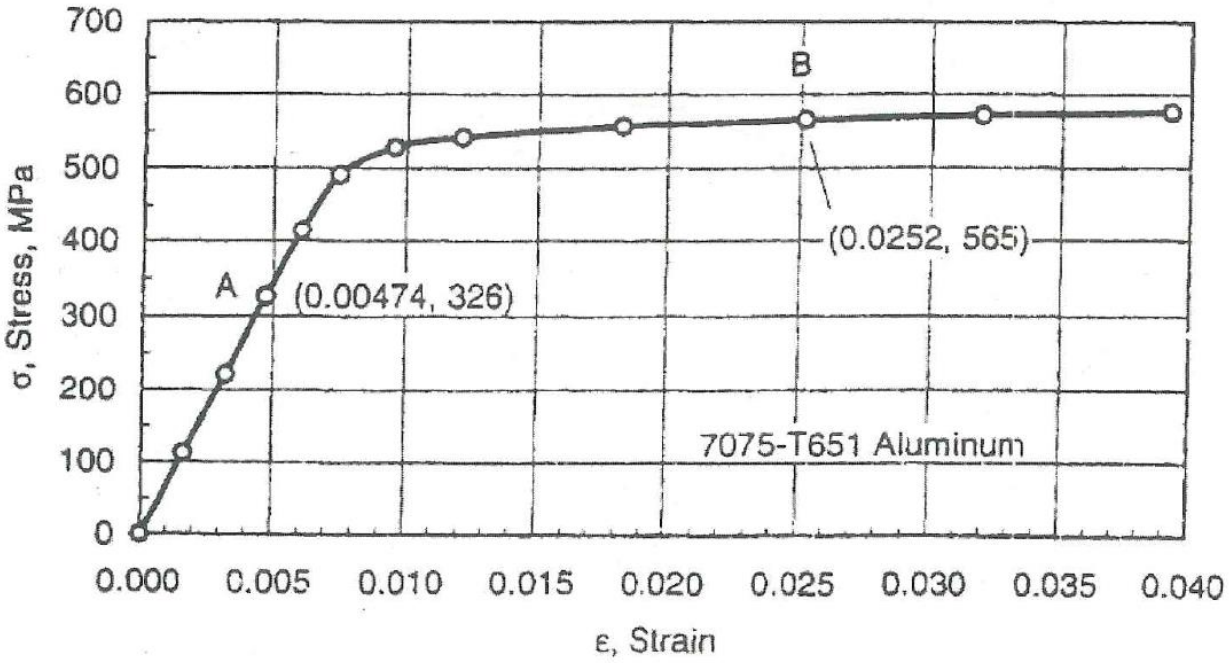
故選(C)

例 4

The initial portion of a tension test on 7075-T651 aluminum. Note that data points A and B are labeled with their stress-strain coordinates.

(a) Obtain the approximate values of the modulus of elasticity and 0.2% offset yield strength.

(b) If a sample of this material 150 mm long is strained to point B and then unloaded, what are the plastic strain and its length remains after unloading?



【成大工科】

【解析】

(a)由圖示中取0.2%平行線相交 $\sigma_y \approx 525 MPa$

(b)

由圖示中

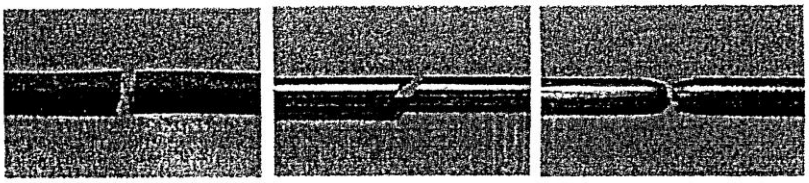
$$\sigma = \mathrm{E}\varepsilon \Rightarrow \mathrm{E} = \frac{326}{0.00474} = 68776.37(MPa)$$

$$\varepsilon_{\mathrm{P}} = \varepsilon - \varepsilon_y = 0.0252 - \frac{565}{68776.37} = 0.017$$

$$L' = L(1+\varepsilon_{\mathrm{P}}) = 150 \times (1+0.017) = 152.55(mm)$$

例 5

附圖之(a)(b)(c)三根桿件試片各受拉伸試驗或者扭轉試驗後失效斷裂，以下敘述何者錯誤？



(a)　　(b)　　(c)

(A)桿件(a)可能為延性材料（ductile material）受扭轉試驗　(B)桿件(b)可能為脆性材料（brittle material）受扭轉試驗　(C)桿件(c)可能為脆性材料受拉伸試驗　(D)桿件(a)可能為脆性材料受拉伸試驗　(E)桿件(c)可能為延性材料受拉伸試驗

【105 台大機械】

【解析】

6. C.

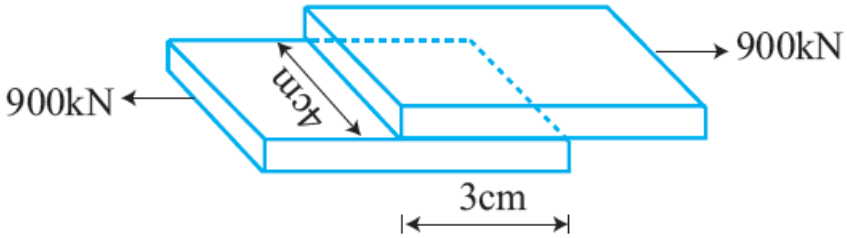
圖(a)脆性材料受拉伸 or 延性材料受扭轉

圖(b)脆性材料受扭轉

圖(c)有頸縮⇒延性材料受拉伸　故(C)有錯誤

例 6

如圖所示之軟鋼方塊其剪割彈性係數G＝90GPa，則其剪應變為若干？



【解析】

$$\tau=\frac{P}{A}=\frac{900}{3\times4}=75\,kN/cm^2=0.75GPa$$

$$\tau=G\times\gamma\rightarrow0.75=90\times\gamma \qquad \therefore\gamma=8.3\times10^{-3}$$

例 7

在一鋁合金(E = 210GPa、ν = 0.3)機械表面上一點 P 受到雙軸應力，由量測所得到之應變為 $\varepsilon_x = +1394\,\mu m/m$，$\varepsilon_y = -660\,\mu m/m$，與 $\gamma_{xy} = +2054\,\mu rad$。試求此點之應力分量 σ_x：　(A)+ 325MPa　(B) + 276MPa　(C)－138MPa　(D)+ 55.8MPa。(96 年經濟部)

承上題，此點之最大剪應力為：　(A)235MPa　(B)213MPa　(C)194MPa　(D)87MPa。(96 年經濟部)

【96 經濟部】

【解析】

$$\varepsilon_x = \frac{\sigma_x}{E} + \left(\frac{-\nu}{E}\right)(\sigma_y)$$

$$\Rightarrow 1394\times10^{-6} = \frac{\sigma_x}{210\times10^9} - \frac{0.3}{210\times10^9}(\sigma_y)\cdots\cdots$$

$$\varepsilon_y = \frac{\sigma_y}{E} + \left(\frac{-\nu}{E}\right)(\sigma_x) \quad \Rightarrow -660\times10^{-6} = \frac{\sigma_y}{210\times10^9} - \frac{0.3\sigma_x}{210\times10^9}\cdots\cdots$$

由　　可得 $\sigma_x = 276\times10^6\,Pa = 276MPa$

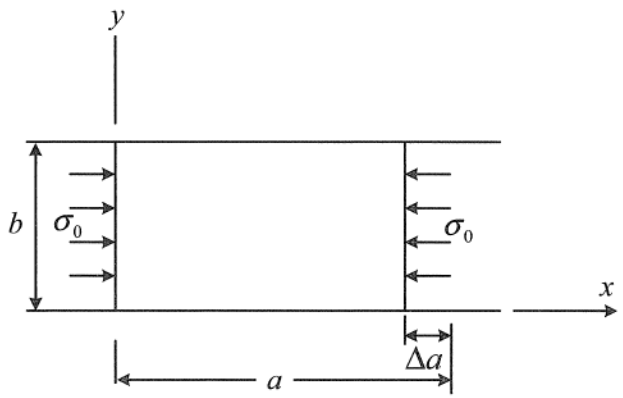
$\sigma_y = -55.8\times10^6\,Pa = -55.8MPa$

$$G\gamma_{xy} = \tau_{xy} \quad G = \frac{E}{2(1+\nu)} = 80.77GPa$$

$$\tau_{xy} = G\gamma = 165.9MPa \quad \tau_{max} = \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2 + (\tau_{xy})^2} = 235.62(MPa)$$

例 8

A block of linear elastic material (E, v) is compressed between two rigid, perfectly smooth surfaces by an applied stress $\sigma_x = -\sigma_o$. The only other nonzero stress is the stress σ_y induced by the retaining surfaces at y = 0 and y = b. Please determine Δa, the chance in the x dimension of the block.



【99 交大機械】

【解析】

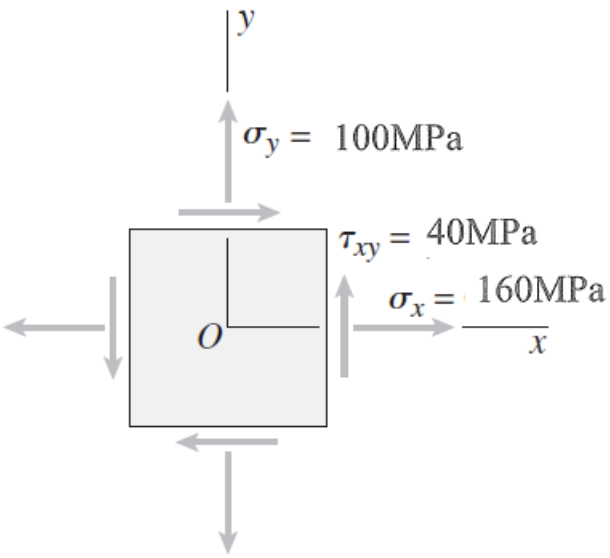
$$\varepsilon_y = \frac{1}{E}(\sigma_y - \nu\sigma_x) = 0 \Rightarrow \sigma_y = -\nu\sigma_0$$

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y) = -\frac{\sigma_0}{E}(1-\nu^2)$$

$$\Delta a = a\varepsilon_x = \frac{a\sigma_0}{E}(1-\nu^2)$$

例 9

The given state of plane strain as shown in Fig. is known to exist at a point in an elastic medium. Given that E=200 GPa, $\nu = 0.3$, determine the principal stresses and strains.



【100 成大機械】

【解析】

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + (\tau_{xy})^2} = 180(Mpa), 80(MPa)$$

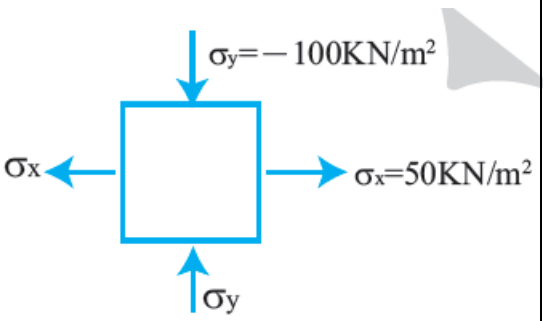
由虎克定律

$$\varepsilon_1 = \frac{1}{E}(\sigma_1 - \upsilon\sigma_2) \Rightarrow \varepsilon_1 = 7.8 \times 10^{-4}$$

$$\varepsilon_2 = \frac{1}{E}(\sigma_2 - \upsilon\sigma_1) \Rightarrow \varepsilon_1 = 1.3 \times 10^{-4}$$

例 10

有一10cm×10cm之薄鋼鈑承受雙軸向應力（biaxial stress）如圖所示，若已知此鋼材之彈性模數為$2\times10^{11}N/m^2$，蒲松比（Poission's ratio）為0.25，則此鈑受力後在x向及y向之變形量為何？又此鈑所受之最大剪應力為何？



【解析】

(1) 由虎克定律

$$\varepsilon_x=\frac{\sigma_x}{E}-\frac{\nu\sigma_y}{E}=\frac{50\times10^3}{2\times10^{11}}-\frac{0.25\times(-100\times10^3)}{2\times10^{11}}=3.75\times10^{-7}$$

$$\varepsilon_x=\frac{\Delta L_x}{L_x}=\frac{\Delta L_x}{10}=3.75\times10^{-7}\Rightarrow\Delta L_x=3.75\times10^{-6}\text{cm(伸長)}$$

同理

$$\varepsilon_y=\frac{\sigma_y}{E}-\frac{\nu\sigma_x}{E}=\frac{(-100\times10^3)}{2\times10^{11}}-\frac{0.25\times(50\times10^3)}{2\times10^{11}}=-5.625\times10^{-7}$$

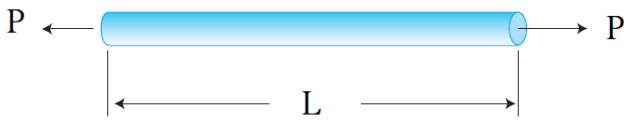
$$\varepsilon_y=\frac{\Delta L_y}{L_y}=\frac{\Delta L_y}{10}\Rightarrow\Delta L_y=-5.625\times10^{-6}\text{ cm(縮短)}$$

(2) 由於$\sigma_x=\sigma_1$，$\sigma_y=\sigma_2$

$$\tau_{max}=\frac{\sigma_1-\sigma_2}{2}=\frac{50\times10^3-(-100\times10^3)}{2}=75(kN/m^2)$$

例 11

A prismatic bar of circular cross section is loaded by tensile forces P (see figure). The bar has length $L = 1.5\text{m}$ and diameter $d = 30\text{mm}$. It is made of aluminum alloy with modulus of elasticity $E = 75\text{GPa}$ and Poisson's ratio $\nu = \frac{1}{3}$. If the bar elongates by $\delta = 3.6mm$, what is the decrease in diameter d? What is the magnitude of the load P?



【中文翻譯】

等截面圓桿受一拉伸負載 P。桿具長度 $L = 1.5\text{m}$ 及直徑 $d = 30\text{mm}$。它為鋁合金所製，具彈性模數 $E = 75\text{GPa}$ 及蒲松比 $\nu = \frac{1}{3}$。若桿之伸長量 $\delta = 3.6mm$，試求直徑減少量 Δd，及受一拉伸負載 P。

【解析】

$$\varepsilon = \frac{\delta}{L} = \frac{3.6\text{ mm}}{1.5\text{ m}} = 0.0024$$

$$\varepsilon' = -\nu\varepsilon = -(\tfrac{1}{3})(0.0024)$$

$$= -0.0008$$

$$\Delta d = \varepsilon' d = (0.0008)(30\,\text{mm}) = 0.024\,\text{mm}$$

$$\sigma = E\varepsilon = (75\,\text{GPa})(0.0024)$$

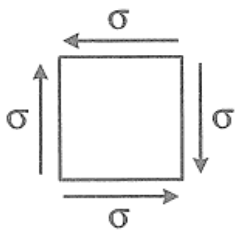
$$= 180\text{ MPa}$$

$$P = \sigma A = (180\text{ MPa})\left(\frac{\pi}{4}\right)(30\text{ mm})^2$$

$$= 127\,\text{kN}$$

例 12

考慮下圖所示受平面應力作用的平板，假設材料的剪切彈性模量為 G，在彈性範圍內，請問平板內的最大主應變為何？



【98 台大機械】

【解析】

$\sigma_1 = \sigma$ 、 $\sigma_2 = -\sigma$

由虎克定律

$$\varepsilon_1 = \frac{1}{E}(\sigma_1 - \upsilon\sigma_2) = \frac{1+\upsilon}{E} = \frac{\sigma}{2G}$$

例 13

At a point on the surface of an alloy steel (E = 210GPa and $\nu = 0.3$) machine part subjected to a biaxial states of stress, the measures strain were $\varepsilon_x = 1394\mu, \varepsilon_y = -660\mu$, $\gamma_{xy} = 2054\mu$ rad. Please determine the principal stresses and the maximum shear stress at the point.

【台大應力】

【解析】

由虎可定律

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \upsilon\sigma_y) \Rightarrow 1394\times10^{-6} = \frac{1}{210\times10^3}(\sigma_x - 0.3\sigma_y)$$

$$\varepsilon_y = \frac{1}{E}(\sigma_y - \upsilon\sigma_x) \Rightarrow -660\times10^{-6} = \frac{1}{210\times10^3}(\sigma_y - 0.3\sigma_x)$$

$$\sigma_y = -55.8(Mpa), \sigma_x = 276(Mpa)$$

$$G = \frac{E}{2(1+\upsilon)} = 80.77(GPa)$$

$$\tau_{xy} = G\gamma_{xy} = 165.9(GPa)$$

最大主應力

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + \tau_{xy}^2} = 344.7\,, -124.5\ (\text{MPa})$$

最大剪應力

$$\tau_{\max} = \frac{\sigma_1 - \sigma_2}{2} = 234.6\ (\text{MPa})$$

例 14

一立方體體積3000cm^{3}丟入水中，若水壓為600MPa，立方體材料之蒲松比（Poisson's ratio）為0.25，彈性係數（modulus of elasticity）為250GPa，試求三軸向應變各為多少？體積應變為多少？變形後的體積為多少？(101關四)

【解析】

(1) $\varepsilon_x=\varepsilon_y=\varepsilon_z=\frac{1}{E}[\sigma_x-\nu(\sigma_y+\sigma_z)]$

其中$\sigma_x=\sigma_y=\sigma_z=P=600(MPa)$

故$\varepsilon_x=\varepsilon_y=\varepsilon_z=\frac{-600}{250\times10^3}\times[1-2\times0.25]$

$=-1.2\times10^{-3}$

(2) $\varepsilon_\forall=\varepsilon_x+\varepsilon_y+\varepsilon_z=-1.2\times10^{-3}\times3=-3.6\times10^{-3}$

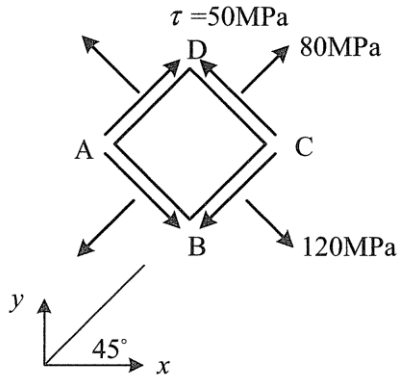
(3) $\forall'=3000-3000\times(3.6\times10^{-3})=2989.2(cm^3)$

例 15

Consider a rectangular plate subjected to 80MPa of tensile stress along edges AB and CD, and 120MPa of tensile stress along edges BC and AD. It is also subjected to 50MPa of shearing stress along the four edges. The direction of the shearing stress is shown in the figure. The plate is made of aluminum with Young's modulus 70GPa and Poisson's ratio 0.35

(a) If the ultimate normal strength of aluminum is 260Mpa and the ultimate shearing strength is 165MPa. Determine the factors of safety of the plate with respect to tensile and shear failure. (10%)

(b) If the original thickness of the plane is 1cm, determine its thickness after the stresses are applied. (15%)



【台大應力】

【解析】

$$\sigma_{av} = \frac{1}{2}(80+120) = 100\ \text{MPa}$$

$$R = \sqrt{20^2 + 50^2} = 53.85\ \text{MPa}$$

$$\sigma_1 = \sigma_{av} + R = 153.85\ \text{MPa}$$

$$\sigma_2 = \sigma_{av} - R = 46.15\ \text{MPa}$$

$$\sigma_3 = 0$$

⇒拉應力的安全因子： $\text{F.S.} = \frac{260}{153.85} = 1.7$

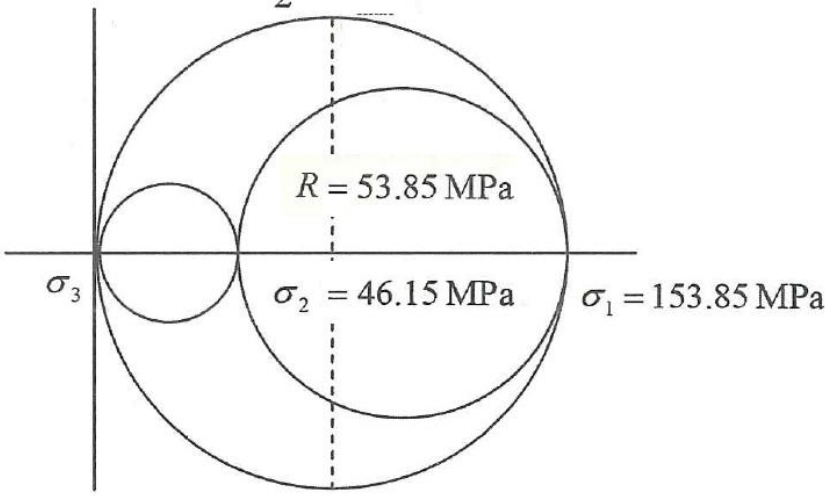
最大剪應力 $\tau_{\max} = \frac{\sigma_1}{2} = 76.9\ (\text{MPa})$

剪應力的安全因子： $\text{F.S.} = \dfrac{165}{76.9} = 2.14$

$$\varepsilon_z = \frac{\sigma_z}{E} - \frac{\nu}{E}(\sigma_x + \sigma_y) = 0 - 0.35 \times \frac{(80 + 120) \times 10^6}{70 \times 10^9} = -0.001$$

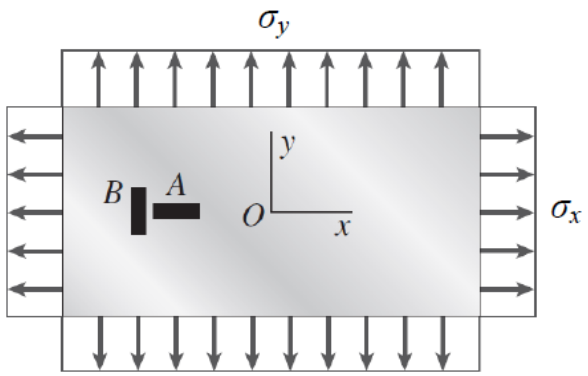
$$t' = t(1 + \varepsilon_z) = 1 \times (1 - 0.001\ ) = 0.999\ (\text{cm})$$

$$\tau_{\max} = \frac{1}{2}(\sigma_1 - \sigma_3) = 76.92\ \text{MPa}$$



例 16

鎂板受雙軸向應力者，受有拉伸應力 $\sigma_x = 24\text{MPa}$ 及 $\sigma_y = 12\text{MPa}$ （ 見圖 ）。板之相對應的應變為及 $\varepsilon_x = 440 \times 10^{-6}$ 、 $\varepsilon_y = 80 \times 10^{-6}$ (shortening)。求材料之蒲松比 ν 及模數 E 。



【解析】

$$\sigma_x = 24\ \text{MPa} \qquad \sigma_y = 12\ \text{MPa}$$
$$\varepsilon_x = 440 \times 10^{-6} \qquad \varepsilon_y = 80 \times 10^{-6}$$

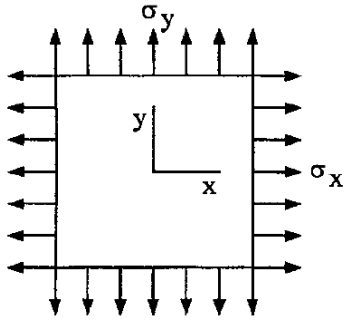
$$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y)$$
$$\varepsilon_y = \frac{1}{E}(\sigma_y - \nu\sigma_x)$$

$$E\,(440 \times 10^{-6}) = 24\ \text{MPa} - \nu\,(12\ \text{MPa})$$
$$E\,(80 \times 10^{-6}) = 12\ \text{MPa} - \nu\,(24\ \text{MPa})$$

$$\nu = 0.35 \qquad E = 45\ \text{GPa}$$

例 17

4. A square plate with uniform thickness t in the z direction is shown below. It is subjected to tri-axial stresses $\sigma_x = 40\,\text{MPa}$, $\sigma_y = 20\,\text{MPa}$ and $\sigma_z = 0\,\text{MPa}$. The measured normal strains are $\varepsilon_x = 17\times10^{-5}$ and $\varepsilon_y = 4\times10^{-5}$. (i) Calculate the Yung's modulus E and the Poisson's ratio ν of the plate. (ii) Calculate the maximum shear stress $(\tau_{xy})_{max}$ in xy plane, the maximum shear stress $(\tau_{yz})_{max}$ in yz plane and the maximum shear stress $(\tau_{xz})_{max}$ in xz plane. (20%)



【106 成大土木】

【解析】

(1) 由廣義虎克定律

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y)$$

$$\Rightarrow 17\times10^{-5}\times E = 40 - \nu\times20 \quad \text{—— ①}$$

$$\varepsilon_y = \frac{1}{E}(\sigma_y - \nu\sigma_x)$$

$$\Rightarrow 4\times10^{-5}\times E = 20 - \nu\times40 \quad \text{—— ②}$$

$$\Rightarrow \begin{cases} E = 200\ \text{GPa} \\ \nu = 0.3 \end{cases}$$

(2) 最大剪應力

σ_x, σ_y, σ_z 恰為主應力

① $(\tau_{xy})_{max} = \frac{40-20}{2} = 10\ (\text{MPa})$

② $(\tau_{yz})_{max} = \frac{20-0}{2} = 10\ (\text{MPa})$

③ $(\tau_{xz})_{max} = \frac{40-0}{2} = 20\ (\text{MPa})$

例 18

一邊長均相等之正立方體承受三軸向應力之作用，若 σ_x = 80MPa 、 σ_y = 50MPa 、 $\sigma_z = -40$ MPa，若正立方體的材料蒲松氏比 $\upsilon = 0.2$ ，則那一個軸向的變形量最小？

【解析】

$$\varepsilon_x = \frac{\sigma_x}{E} - \upsilon\frac{\sigma_y}{E} - \upsilon\frac{\sigma_z}{E}$$

$$= \frac{80}{E} - 0.2\times\frac{50}{E} - 0.2\times\frac{-40}{E} = \frac{78}{E}$$

$$\varepsilon_y = \frac{\sigma_y}{E} - \upsilon\frac{\sigma_x}{E} - \upsilon\frac{\sigma_z}{E}$$

$$= \frac{50}{E} - 0.2\times\frac{80}{E} - 0.2\times\frac{-40}{E} = \frac{42}{E}$$

$$\varepsilon_z = \frac{\sigma_z}{E} - \upsilon\frac{\sigma_x}{E} - \upsilon\frac{\sigma_y}{E}$$

$$= \frac{-40}{E} - 0.2\times\frac{80}{E} - 0.2\times\frac{50}{E} = \frac{-66}{E}$$

由上可知，ε_y 變形量最小(不管張應力或壓應力)。

例 19

有一材料之體積 $V = 2000\ mm^3$、蒲松氏比 $\upsilon = 0.15$、彈性係數 $E = 70GPa$，承受三軸向應力作用，其應力分別為 $\sigma_x = 40\ MPa$、$\sigma_y = 30\ MPa$、$\sigma_z = -20\ MPa$，則此材料之體積變化量若干？

【解析】

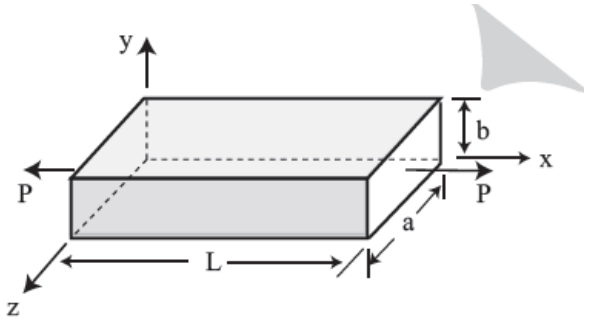
$$\varepsilon_V = \frac{1-2\upsilon}{E}(\sigma_x + \sigma_y + \sigma_y)$$

$$= \frac{1-2\times 0.15}{70\times 10^3}\times(40+30-20) = 0.0005$$

$$\Delta V = \varepsilon_V V = 0.0005\times 2000 = 1\ mm^3$$

例 20

一塊四方體寬a，深b，長L承受軸向拉力P如圖所示。隨著張力，b及L的尺寸分別變成1.999in.及10.02in.。試計算(1)蒲松比。(2)彈性模數。(3)尺寸a的最後值。(4)切向彈性模數。已知：a=3in.，b=2in.，L=10in.，P=100kips



【解析】

由 $\sigma_x = \dfrac{100(10^3)}{3(2)} = 16.67\,\text{ksi}$

(1) $\varepsilon_x = \dfrac{0.02}{10} = 2\times10^{-3}\,\mu \quad \varepsilon_y = -\dfrac{0.001}{2} = \underline{-5\times10^{-4}\,\mu} \quad \upsilon = \left|\dfrac{500}{2000}\right| = \underline{0.25}$

(2) $\mathrm{E} = \dfrac{\sigma_x}{\varepsilon_x} = \dfrac{16.67(10^3)}{2\times10^{-3}} = \underline{8.335(10^6)\ \text{psi}}$

(3) $\varepsilon_z = -\dfrac{\nu\ \sigma_x}{\mathrm{E}}(-0.25)\times\dfrac{-16.67\times10^3}{8.335\times10^6} = 5\times10^{-4}$

$\Delta a = 500(10^{-4})3 = 1.5(10^{-3})\ \text{in.} \ ; \quad a' = a + \Delta a = 2.9985(\text{in})$

(4) $\mathrm{G} = \dfrac{8.335\left(10^6\right)}{2(1+0.25)} = 3.334\left(10^6\right)\ \text{psi}$

例 21

有一材料之體積V＝2000mm^3、蒲松氏比 μ ＝015、彈性係數E＝70GPa，承受三軸向應力作用，其應力分別為 σ_x=40MPa、σ_y=30MPa、σ_z=－20MPa，則此材料之體積變化量若干？

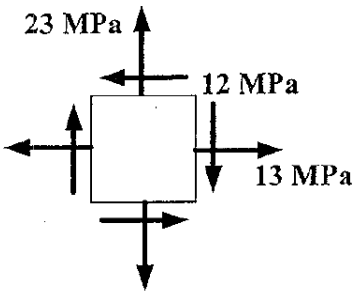
【解析】

$$\varepsilon_V = \frac{1-2\mu}{E}(\sigma_x + \sigma_y + \sigma_y) = \frac{1-2\times 0.15}{70\times 10^3}\times(40+30-20) = 0.0005$$

$$\Delta V = \varepsilon_V V = 0.0005\times 2000 = 1\ \text{mm}^3$$

例 22

The in-plane stress state of a material point under plane strain condition is shown. Given that $E = 100$ GPa, $\nu = 1/3$ and $\sigma_Y = 70$ MPa, determine
(a) the principal stresses, (10 points)
(b) the principal strains. (10 points)



【106 成大機械】

【解析】

(a)平面主應力

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{(\frac{\sigma_x - \sigma_y}{2})^2 + (\tau_{xy})^2} = \frac{13 + (23)}{2} \pm \sqrt{(\frac{13 - (23)}{2})^2 + (-12)^2} = 31{,}5(MPa)$$

(b) 平面主應變

由虎克定律

$$\varepsilon_1 = \frac{1}{E}(\sigma_1 - \upsilon\sigma_2) \Rightarrow \varepsilon_1 = 2.93 \times 10^{-4}$$

$$\varepsilon_2 = \frac{1}{E}(\sigma_2 - \upsilon\sigma_1) \Rightarrow \varepsilon_2 = -5.3 \times 10^{-5}$$

例 23

A cube of cast iron with sides of length 75 mm is tested in a laboratory under triaxial stress. Gages mounted on the specimen show that the compressive strains are $\varepsilon_x = -350\times10^{-6}$ and $\varepsilon_y = \varepsilon_z = -65\times10^{-6}$. Please determine the following quantities: (a) the normal stress $\sigma_x, \sigma_y, \sigma_z$ acting on the x, y, and z faces of the cube. (b) The maximum shear stress in the material. (Assume $E = 96GPa \quad v = 0.25$)

【98 交大機械】

【解析】

由虎克定律

$$\varepsilon_x = \frac{1}{E}[\sigma_x - \upsilon(\sigma_y + \sigma_z)] \Rightarrow -350\times10^{-6} = \frac{1}{96000}[\sigma_x - 0.25(\sigma_y + \sigma_z)]$$

$$\varepsilon_y = \frac{1}{E}[\sigma_y - \upsilon(\sigma_x + \sigma_z)] \Rightarrow -65\times10^{-6} = \frac{1}{96000}[\sigma_y - 0.25(\sigma_x + \sigma_z)]$$

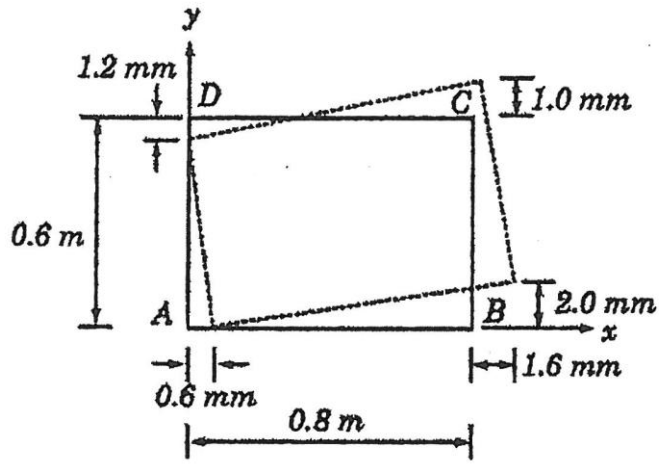
$$\varepsilon_z = \frac{1}{E}[\sigma_z - \upsilon(\sigma_x + \sigma_y)] \Rightarrow \varepsilon - 65\times10^{-6} = \frac{1}{96000}[\sigma_z - 0.25(\sigma_x + \sigma_y)]$$

$$\sigma_y = -23.4(Mpa), \sigma_x = -45.3(Mpa), \sigma_z = -23.4(Mpa)$$

最大剪應力 $\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2} = 10.95(MPa)$

例 24

A 0.8-m by 0.6-m rectangle ABCD is drawn on a thin plate prior to loading. Subsequent to loading, the deformed geometry is shown by the dash lines. The material's modulus of elasticity $E = 210$ GPa, and Poisson's ratio $\nu = 0.3$. (1) Determine the components of plane strain at point A. (10%) (2) Determine the in-plane principal strains at point A and the directions in which they act. (10%) (3) Determine the principal stresses at point A. (10%)



【成大土木】

【解析】

$$\varepsilon_x = \frac{1}{800} = 1.25 \times 10^{-3}$$

$$\varepsilon_y = -\frac{1.2}{600} = -2 \times 10^{-3}$$

$$\gamma_{xy} = \frac{2}{800} - \frac{0.6}{600} = 1.5 \times 10^{-3}$$

$$\theta_P = \frac{1}{2}\tan^{-1}\left(\frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y}\right) = \begin{cases} 12.39^o \\ 102.39^o \end{cases}$$

$$\varepsilon_P = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2} = \begin{cases} 1.415 \times 10^{-3} \ (= \varepsilon_a) \\ -2.165 \times 10^{-3} \ (= \varepsilon_b) \end{cases}$$

$$\varepsilon_a = \frac{\sigma_a}{E} - \nu\frac{\sigma_b}{E}$$

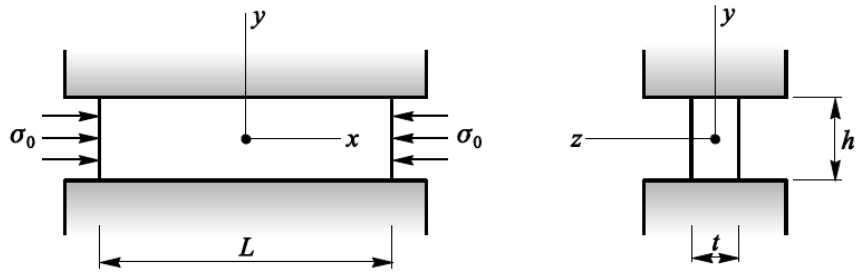
$$\varepsilon_b = \frac{\sigma_b}{E} - \nu\frac{\sigma_a}{E}$$

$$\sigma_a = \frac{E}{1-\nu^2}(\varepsilon_a + \nu\varepsilon_b) = 176.65\ MPa$$

$$\sigma_b = \frac{E}{1-\nu^2}(\varepsilon_b + \nu\varepsilon_a) = -401.65\ MPa$$

例 25

圖中一長方體之材料(長度為 L 高度為 h 厚度為 t)置於距離為 h 的上下兩個固定剛性牆壁間，今在左右兩端施加均勻之壓應力σ_0。試求(a)材料在上下兩面所受之壓力 p(設材料與剛性壁間之摩擦忽略不計)，(b)材料之膨脹率 e，(c)材料之應變能密度。設彈性模數為 E，蒲松比為ν。



【108 台科營建】

【解析】

$$\sigma_x=-\sigma_0\ ,\ \sigma_y=-p\ ,\ \sigma_z=0\ ,\ \varepsilon_y=0$$

$$\text{(a)}\ \varepsilon_y=\frac{\sigma_y-\nu\sigma_x}{E}=\frac{-p-\nu(-\sigma_0)}{E}=0\ ,\ p=\nu\sigma_0$$

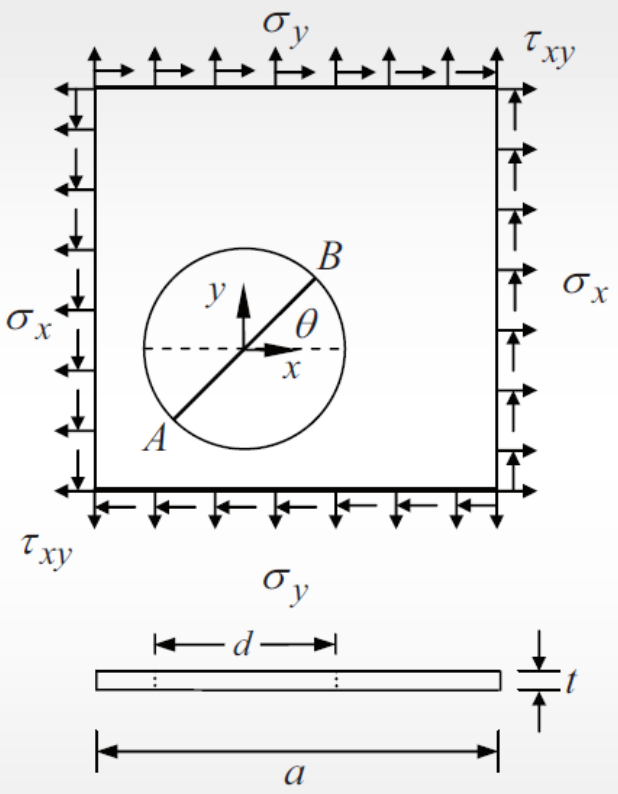
$$\varepsilon_x=\frac{\sigma_x-\nu\sigma_y}{E}=\frac{-\sigma_0-\nu(-\nu\sigma_0)}{E}=-\frac{\sigma_0}{E}(1-\nu^2)$$

$$\text{(b)}\ e=\frac{1-2\nu}{E}(\sigma_x+\sigma_y)=\frac{1-2\nu}{E}(-\sigma_0-\nu\sigma_0)=-\frac{\sigma_0}{E}(1-2\nu)(1+\nu)$$

$$\text{(c)}\ u=\frac{1}{2}(\sigma_x\varepsilon_x+\sigma_y\varepsilon_y)=\frac{1}{2}(-\sigma_0)\left[-\frac{\sigma_0}{E}(1-\nu^2)\right]=\frac{\sigma_0^2}{2E}(1-\nu^2)$$

範例 26

直徑為 d=4cm 之圓為如圖所示之等向均質彈性正方形薄板（Isotropic homogeneous elastic thin plate）的一部分，線段 $\overline{AB}$ 為該圓的直徑，$\overline{AB}$ 在變形前與 x 軸的夾角為 θ，該薄板的邊長為 a=20cm、厚度為 t=0.1cm、楊氏模數（Young's modulus）為 E=200GPa，包松比（Poisson's ratio）為 ν=0.3。該薄板受一均勻平面應力（Plane stress）$\sigma_x = -200\text{MPa}$、$\sigma_y = 300\text{MPa}$、$\tau_{xy} = 200\text{MPa}$ 作用而變形，令圓的直徑 $\overline{AB}$ 變形後的長度為 d*，$\overline{AB}$ 的伸長量 $\delta = d^* - d$，試求當 θ 為多少度時，δ 有最大值及 δ 的最大值為多少？（102 地三）



【解析】

解 (1) 由廣義虎克定律求應變

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \upsilon\sigma_y) = -1450\mu$$

$$\varepsilon_y = \frac{1}{E}(\sigma_y - \upsilon\sigma_x) = -1800\mu$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G} = \frac{2(1+\upsilon)\tau_{xy}}{E} = 2600\mu$$

(2) 應變轉換公式

$$\varepsilon_\theta = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = 175 - 1625\cos 2\theta + 1300\sin 2\theta$$

$$\delta = \varepsilon_\theta d = (175 - 1625\cos 2\theta + 1300\sin 2\theta)d = d^* - d$$

$$\frac{d\delta}{d\theta} = (+2\times 1625\sin 2\theta + 2\times 1300\cos 2\theta)d = 0 \Rightarrow 2\theta = \tan^{-1}(0.8) = -38.66°$$

例 27

4. (25%) Consider a metal pipe of length L=1.2 m with an outside diameter d_2=5cm and inner diameter d_1=3cm. One side of this pipe is fixed on the floor. The top side of this pipe has a compressive force $F = 4\pi \times 10^4$ N applied on its surface. The elasticity and poisson's ratio of this metal is E = 200 GPa and ν = 0.3, respectively.

Determine the following quantities for this pipe:

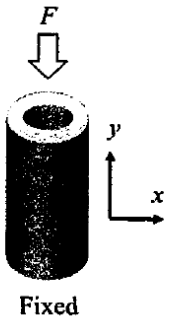
(1) The axial normal stress σ_y , strain ε_y and the shorting δ (9%)

(2) the lateral strain ε_x (3%)

(3)The change of outer diameter Δd_2 and inner Δd_1 (6%)

(4) the increase in wall thickness Δt (3%)

(5) the dilation e (4%)



【106 台大應力】

【解析】

<Sol>: (1) $\sigma_y = -\frac{F}{A} = -\frac{4\pi \times 10^4}{\frac{\pi}{4}(50^2 - 30^2)} = -100\ (MPa)$

$\varepsilon_y = \frac{\sigma_y}{E} = \frac{-100}{200 \times 10^3} = -5 \times 10^{-4}$ ← 應變沒有單位

$\delta = \varepsilon_y \times L = (-5 \times 10^{-4}) \times 1200 = \ominus 0.6\ (mm)$ 縮短

(2) $\varepsilon_x = -\nu \varepsilon_y$ ← ∵是單軸向負載才可以用

$= -0.3 \times (-5 \times 10^{-4})$

$= 1.5 \times 10^{-4}$

不是的話就要用廣義虎克定律去找

(3) $\Delta d_2 = \varepsilon_x \times d_2 = 7.5 \times 10^{-3}\ (mm)$ ← 用環向應變來算

$\Delta d_1 = \varepsilon_x \times d_1 = 4.5 \times 10^{-3}\ (mm)$

(4) $\Delta t = \frac{\Delta d_2 - \Delta d_1}{2} = 1.5 \times 10^{-3}\ (mm)$

Δt = 側向應變 × t (ε_x 或 ε_z)

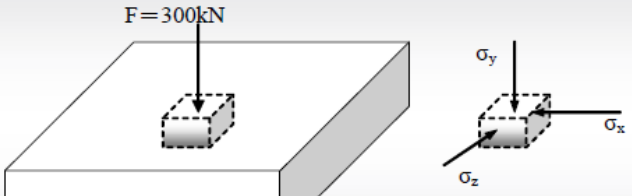
(5) $e = \varepsilon_v = \varepsilon_x + \varepsilon_y + \varepsilon_z$ ← ($\varepsilon_x = \varepsilon_z$, ∵都是側向)

$= (1 - 2\nu)\varepsilon_y$

$= \ominus 2 \times 10^{-4}$

例 28

邊長為 0.1m 的銅立方塊，無間隙地放入變形可略去不計的剛性凹槽中，已知銅的彈性模量 E=100GPa，蒲松比 ν=0.34。當銅塊受到了 F=300kN 的均佈壓力作用時，試求銅塊的三個主應力的大小。



【中正機械】

【解析】

1. 銅塊 y 方向的壓應力為 $\sigma_y = -\frac{F}{A} = -\frac{300\times10^3}{0.1^2} = -30\text{MPa}$ (壓為負)

2. 銅塊受到 y 方向壓縮將產生三軸向的變形，但由於剛性凹槽壁的約束，使得銅塊在 x，z 方向的應變等於零。於是，在銅塊與凹槽壁接觸面將產生均勻的壓應力 σ_x，σ_z，按廣義虎克定律公式，可得

$$\varepsilon_x = \frac{1}{E}\left[\sigma_x - \nu\left(\sigma_y + \sigma_z\right)\right] = 0$$

$$\varepsilon_z = \frac{1}{E}\left[\sigma_z - \nu\left(\sigma_x + \sigma_y\right)\right] = 0$$

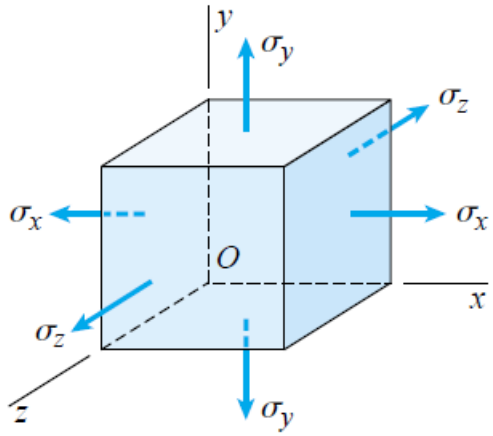
聯解兩式，可得 $\sigma_x = \sigma_z = \frac{\nu}{1-\nu}\sigma_y = \frac{0.34}{1-0.34}(-30) = -15.5\text{MPa}$

3. 按主應力的代數值順序排列，該銅塊的主應力為：

$$\sigma_1 = \sigma_2 = -15.5\text{MPa}, \quad \sigma_3 = -30\text{MPa}$$

例 29

An element of aluminum in triaxial stress (see figure) is subjected to stresses $\sigma_x = 36\ \text{MPa}$ (tension), $\sigma_y = -30\ \text{MPa}$ (compression), and $\sigma_z = -21\ \text{MPa}$ (compression). It is also known that the normal strains in the x and y directions are $\varepsilon_x = 713.8\times10^{-6}$ (elongation) and $\varepsilon_y = -502.3\times10^{-6}$ (shortening). What is the bulk modulus K for the aluminum?



【104 台大土木】

【中文翻譯】

受三軸應力之鋁塊（ 見圖 ），受有應力 $\sigma_x = 36\ \text{MPa}$（ 拉伸 ）， $\sigma_y = -30\ \text{MPa}$（ 壓縮 ），以及 $\sigma_z = -21\ \text{MPa}$（ 壓縮 ）。也知道了在 x 及 y 方向之正向應變為 $\varepsilon_x = 713.8\times10^{-6}$（ 拉長 ）及 $\varepsilon_y = -502.3\times10^{-6}$（ 縮短 ）。鋁的體積模數 K 為何？

【解析】

由虎克定律

$$\varepsilon_x = \frac{1}{E}[\sigma_x - \upsilon(\sigma_y + \sigma_z)] \Rightarrow 713.8\times10^{-6} = \frac{1}{E}[36 - \upsilon(-30 + (-21))]$$

$$\varepsilon_y = \frac{1}{E}[\sigma_y - \upsilon(\sigma_x + \sigma_z)] \Rightarrow -502.3\times10^{-6} = \frac{1}{E}[-33 - \upsilon(36 + (-21))]$$

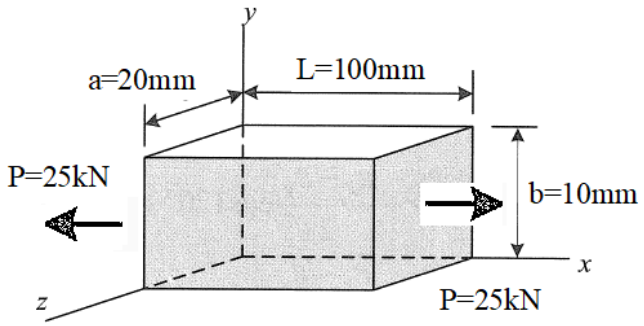
求得 $E = 75652.4(MPa)$ 、 $\upsilon = 0.33$

$$\varepsilon_z = \frac{1}{E}[\sigma_z - \upsilon(\sigma_x + \sigma_y)] \Rightarrow \varepsilon_z = -290.7\times10^{-6}$$

$$K = \frac{E}{3(1-2\upsilon)} = 74169(MPa)$$

例 30

圖所示的方塊受軸向力 P。試計算軸向應變。已知：P=25kN，a=20mm，b=10mm，L=100mm，E=70Gpa，ν =0.3。假設：方塊限制不得於 y-及 z-方向收縮。



【解析】

已知

$$\varepsilon_y = \varepsilon_z = 0 \qquad \sigma_x = \frac{25\left(10^3\right)}{20\times 10\left(10^{-6}\right)} = 125 \text{ MPa}$$

所以

$$\varepsilon_y = 0 = \tfrac{1}{E}\left[\sigma_y - \nu\left(\sigma_x + \sigma_z\right)\right] \qquad \text{(a)}$$

$$\varepsilon_z = 0 = \tfrac{1}{E}\left[\sigma_z - \nu\left(\sigma_x + \sigma_y\right)\right] \qquad \text{(b)}$$

$$\varepsilon_x = \tfrac{1}{E}\left[\sigma_x - \nu\left(\sigma_y + \sigma_z\right)\right] \qquad \text{(c)}$$

式 (a) 及 (b) 變成

$$\sigma_y - \nu\sigma_z = \nu\sigma_x \qquad \text{(a')}$$

$$\sigma_z - \nu\sigma_y = \nu\sigma_x \qquad \text{(b')}$$

加上 $\nu\left(\sigma_y + \sigma_z\right) = 2\nu^2\sigma_x/\left(1-\nu\right)$. 於是式 (c)：

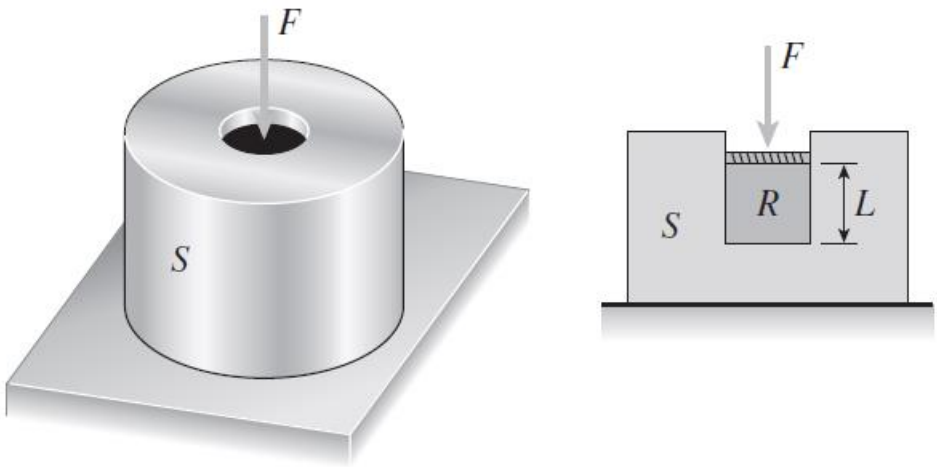
$$\varepsilon_x = \frac{1-\nu-2\nu^2}{1-\nu}\frac{\sigma_x}{E}$$

代入數字：

$$\varepsilon_x = \frac{1-0.3-0.18}{0.7}\frac{125\left(10^6\right)}{70\left(10^9\right)} = \underline{1327\mu}$$

例 31

A rubber cylinder R of length L and cross-sectional area A is compressed inside a steel cylinder S by a force F that applies a uniformly distributed pressure to the rubber (see figure). (a) Derive a formula for the lateral pressure p between the rubber and the steel. (Disregard friction between the rubber and the steel, and assume that the steel cylinder is rigid when compared to the rubber.) (b) Derive a formula for the shortening δ of the rubber cylinder.



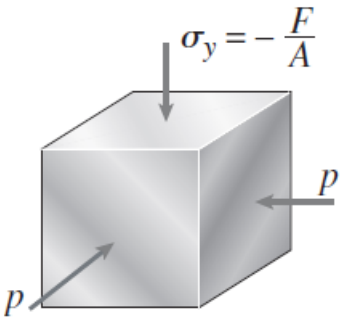
【106 台大土木】

【中文翻譯】

橡膠圓柱 R，長 L，橫截面面積為 A，在鋼圓柱 S 內受一力 F 之壓縮；力 F 產生均勻的分佈的壓力於橡膠（ 如圖 ）。(a) 導出介於橡膠及鋼之間的側向壓力 P 之公式。(忽略橡膠與鋼之間的摩擦力，並假設與橡膠相比，鋼是剛性的)。(b) 導出橡膠圓柱之縮短量 δ 之公式。

【解析】

(a)



$$\sigma_x = -p \qquad \sigma_y = -\frac{F}{A}$$
$$\sigma_z = -p$$
$$\varepsilon_x = \varepsilon_z = 0$$

$$\varepsilon_x = \frac{\sigma_x}{E} - \frac{\nu}{E}(\sigma_y + \sigma_z)$$

$$0 = -p - \nu\left(-\frac{F}{A} - p\right)$$

$$p = \frac{\nu}{1-\nu}\left(\frac{F}{A}\right)$$

(b)

$$\varepsilon_y = \frac{\sigma_y}{E} - \frac{\nu}{E}(\sigma_z + \sigma_x) = -\frac{F}{EA} - \frac{\nu}{E}(-2p)$$

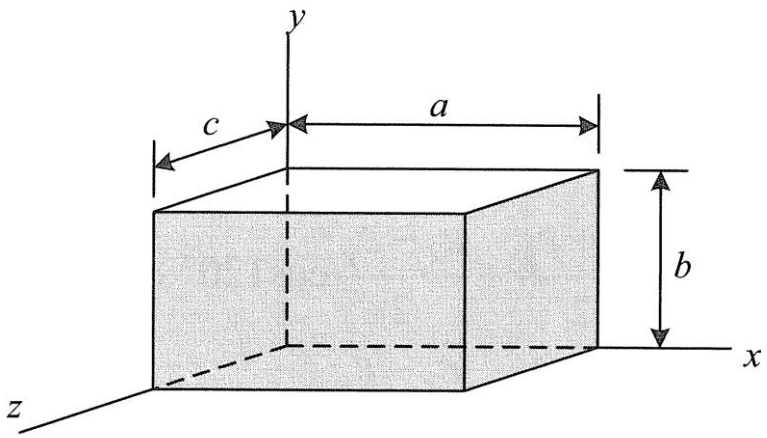
$$\varepsilon_y = \frac{F}{EA}\frac{(1+\nu)(-1+2\nu)}{1-\nu}$$

$$\delta = -\varepsilon_y L$$

$$\delta = \frac{(1+\nu)(1-2\nu)}{(1-\nu)}\left(\frac{FL}{EA}\right)$$

例 32

An element of aluminum in the form of a rectangular parallelepiped (see figure) of dimensions a=6.0in.,b=4.0in, andc=3.0in.is subjected to triaxial stresses $\sigma_x = 12000\text{psi}$, $\sigma_y = -4000\text{psi}$, $\sigma_z = -1000\text{psi}$ acting on the x, y, and z faces, respectively. Determine the following quantities: (a) the maximum shear stress $\tau_{\max}$ in the material; (b) the changes Δa 、 Δb and Δc in the dimensions of the element; (c) the change ΔV in the volume; and (d) the strain energy U stored in the element. (Assume E =10,400 ksi and $\nu = 0.33$)



【成大航太類題】【交大機械類題】

【中文翻譯】

一塊鋁形成一矩形六面體（ 見圖 ），尺寸為a=6.0 in.、b = 4.0 in、and c = 3.0 in，受有三軸應力 $\sigma_x = 12000\text{psi}$ 、 $\sigma_y = -4000\text{psi}$ 及 $\sigma_z = -1000\text{psi}$ ，分別作用在 x 、 y 及 z 平面上。求如下量：(a) 材料之最大剪應力 $\tau_{\max}$ ；(b) 鋁塊之尺寸的變化 Δa 、 Δb 及 Δc ；(c) 體積的改變量 ΔV ；及 (d) 貯於鋁塊之應變能 U 。(假設 $E = 70\text{ GPa}$ 及 $\nu = 0.33$)。

【解析】

$$\sigma_x = 12{,}000 \text{ psi} \quad \sigma_y = -4{,}000 \text{ psi}$$
$$\sigma_z = -1{,}000 \text{ psi}$$
$$a = 6.0 \text{ in.} \quad b = 4.0 \text{ in.} \quad c = 3.0 \text{ in.}$$
$$E = 10{,}400 \text{ ksi} \quad \nu = 0.33$$

(a)

$$\sigma_1 = 12{,}000 \text{ psi} \quad \sigma_2 = -1{,}000 \text{ psi}$$
$$\sigma_3 = -4{,}000 \text{ psi}$$
$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = 8{,}000 \text{ psi}$$

(b)

$$\varepsilon_x = \frac{\sigma_x}{E} - \frac{\nu}{E}(\sigma_y + \sigma_z) = 1312.5 \times 10^{-6}$$
$$\varepsilon_y = \frac{\sigma_y}{E} - \frac{\nu}{E}(\sigma_z + \sigma_x) = -733.7 \times 10^{-6}$$
$$\varepsilon_z = \frac{\sigma_z}{E} - \frac{\nu}{E}(\sigma_x + \sigma_y) = -350.0 \times 10^{-6}$$

$$\Delta a = a\varepsilon_x = 0.0079 \text{ in.}$$
$$\Delta b = b\varepsilon_y = -0.0029 \text{ in.}$$
$$\Delta c = c\varepsilon_z = -0.0011 \text{ in.}$$

(c)

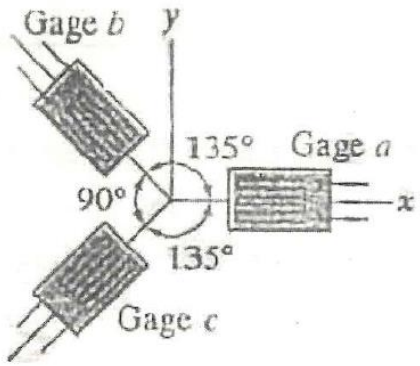
$$e = \frac{1 - 2\nu}{E}(\sigma_x + \sigma_y + \sigma_z) = 228.8 \times 10^{-6}$$
$$V = abc$$
$$\Delta V = e\,(abc) = 0.0165 \text{ in.}^3$$

(d)

$$u = \frac{1}{2}(\sigma_x\varepsilon_x + \sigma_y\varepsilon_y + \sigma_z\varepsilon_z)$$
$$= 9.517 \text{ psi}$$
$$U = u\,(abc) = 685 \text{ in.-lb}$$

例 33

At a point on the surface of a steel machine component, the strain rosette measured the following strains. $\varepsilon_a = 700\mu$, $\varepsilon_b = 560\mu$, and $\varepsilon_c = -280\mu$. Letting $E = 210\,\text{GPa}$ and $\nu = 0.3$, determine the stress components σ_x, σ_y and τ_{xy} at the rosette location.



【101 台大土木】

【解析】

$\varepsilon_x = 700\mu = \varepsilon_a$

$$\varepsilon_{135^\circ} = \varepsilon_b = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = 560\mu$$

$$\Rightarrow \frac{700 + \varepsilon_y}{2} + \frac{700 - \varepsilon_y}{2}\cos 2\times(135^\circ) + \frac{\gamma_{xy}}{2}\sin 2\times(135^\circ) = 560\mu$$

同理

$$\varepsilon_{-135^\circ} = \varepsilon_b = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = -280\mu$$

$$\Rightarrow \frac{700 + \varepsilon_y}{2} + \frac{700 - \varepsilon_y}{2}\cos 2\times(-135^\circ) + \frac{\gamma_{xy}}{2}\sin 2\times(-135^\circ) = -280\mu$$

由上兩式可解得

$\varepsilon_y = -420\mu$ 、 $\gamma_{xy} = -840\mu$

由虎克定律

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \upsilon\sigma_y) \Rightarrow 700\times10^{-6} = \frac{1}{210\times10^3}(\sigma_x - 0.3\sigma_y)$$

$$\varepsilon_y = \frac{1}{E}(\sigma_y - \upsilon\sigma_x) \Rightarrow -420\times10^{-6} = \frac{1}{210\times10^3}(\sigma_y - 0.3\sigma_x)$$

$$\sigma_y = -48.46(Mpa), \sigma_x = 132.42(Mpa)$$

$$G = \frac{E}{2(1+\upsilon)} = 80.77(GPa)$$

$$\tau_{xy} = G\gamma_{xy} = -74.57(GPa)$$

例 34

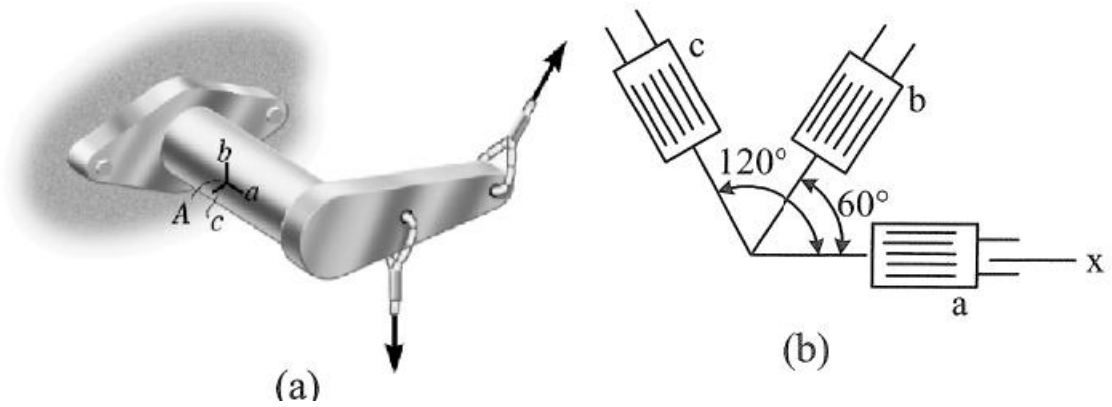
The state of strain at point A on the bracket in Figure (a) is measured using the strain rosette shown in Figure (b). The readings from the gages give

$\epsilon_a = 60(10^{-6}),\quad \epsilon_b = 135(10^{-6})\quad \epsilon_c = 264(10^{-6})$

and The material's modulus of elasticity $E = 200\,\text{GPa}$, and Poisson's ratio $\nu = 0.3$.

(1) Determine the in-plane principal strains at the point and the directions in which they act.

(2) Determine the principal stresses at point A.



【成大土木】

【中文翻譯】

在圖 (a) 中托架上 A 點的應變狀態是用圖 (b) 中所示應變花式量測得。由於負載，量規上讀數為 $\epsilon_a = 60(10^{-6})$, $\epsilon_b = 135(10^{-6})$ 及 $\epsilon_c = 264(10^{-6})$ 試求此點的平面主應變及其作用方向。

【解析】

(一)

$$\varepsilon_x = \varepsilon_a = 60\mu$$

$$\varepsilon_b = 135\mu = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 120° + (\frac{\gamma_{xy}}{2})\sin 120°$$

$$\varepsilon_c = 264\mu = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 240° + (\frac{\gamma_{xy}}{2})\sin 240°$$

$$\varepsilon_y = 246\mu\,,\ \gamma_{xy} = -149\mu$$

(二)

利用莫爾圓可求出平面主應變

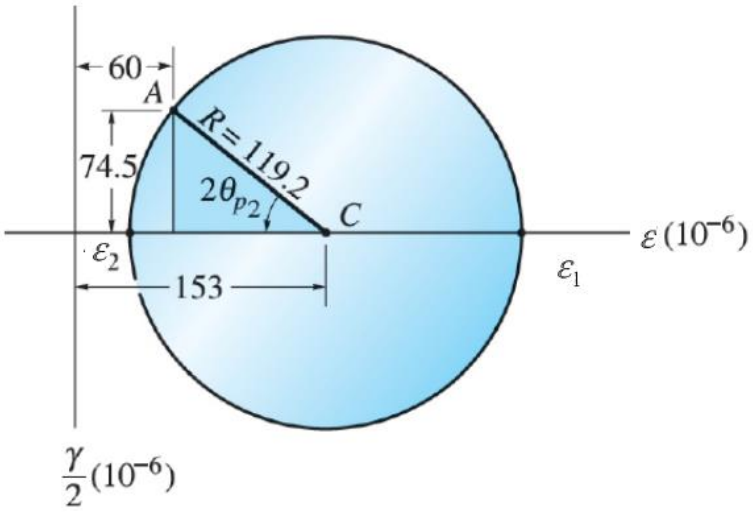
圓上參考點為 $A(60(10^{-6}), -74.5(10^{-6}))$

圓心 C 在 ε 軸上位於 $\varepsilon_{avg} = 153(10^{-6})$

$$R=[\sqrt{(153-60)^2+(74.5)^2}](10^{-6})=119.2(10^{-6})$$

$$\theta_p=\frac{1}{2}\tan^{-1}(\frac{\gamma_{xy}}{\varepsilon_x-\varepsilon_y})=19.3^\circ,-70.7^\circ$$

$$\varepsilon_{1,2}=\frac{\varepsilon_x+\varepsilon_y}{2}\pm\sqrt{(\frac{\varepsilon_x-\varepsilon_y}{2})^2+(\frac{\gamma_{xy}}{2})^2}$$
$$=272\mu\,(-70.7^\circ),33.8\mu\,(19.3^\circ)$$



(三)

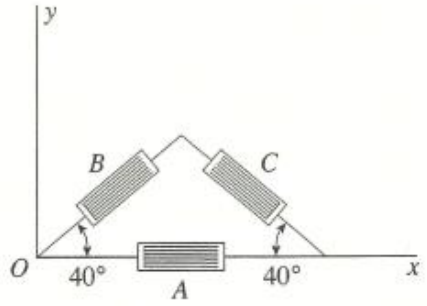
主應力

$$\sigma_1=E\frac{(\varepsilon_1+\nu\varepsilon_2)}{1-\nu^2}=62\,(\text{MPa})$$

$$\sigma_2=E\frac{(\varepsilon_2+\nu\varepsilon_1)}{1-\nu^2}=25.4\,(\text{MPa})$$

例 35

The strains on the surface of an experimental device made of pure aluminum ($E = 70\,\text{GPa}$ ， $\nu = 0.33$) and tested in a space shuttle were measured by means of strain gages. The gages were oriented as shown in the figure, and the measured strains were $\varepsilon_a = 1100 \times 10^{-6}$， $\varepsilon_b = 1496 \times 10^{-6}$, and $\varepsilon_c = -39.44 \times 10^{-6}$, What is the stress σ_x in the x direction?



【105 台大土木大地組】

【中文翻譯】

純鋁 ($E = 70\,\text{GPa}$ ， $\nu = 0.33$) 製實驗裝置表面上之應變，藉應變規在太空梭內做測試。規的安排如圖所示，且量得的應變為 $\varepsilon_a = 1100 \times 10^{-6}$， $\varepsilon_b = 1496 \times 10^{-6}$, 及 $\varepsilon_c = -39.44 \times 10^{-6}$, 。在 x 方向之應力 σ_x 為何？

【解析】

$$\theta = 0° \quad \varepsilon_A = 1100 \times 10^{-6}$$
$$\theta = 40° \quad \varepsilon_B = 1496 \times 10^{-6}$$
$$\theta = 140° \quad \varepsilon_C = -39.44 \times 10^{-6}$$

$$\theta = 0°: \quad \varepsilon_x = \varepsilon_A = 1100 \times 10^{-6}$$

$$\theta = 40°:$$

$$\varepsilon_{x_1} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = \varepsilon_B = 1496 \times 10^{-6}$$

$$0.41318\varepsilon_y + 0.49240\gamma_{xy} = 850.49 \times 10^{-6}$$

$$\theta = 140°:$$

$$\varepsilon_{x_1} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = \varepsilon_C = -39.44 \times 10^{-6}$$

$$0.41318\varepsilon_y - 0.49240\gamma_{xy} = -684.95 \times 10^{-6}$$

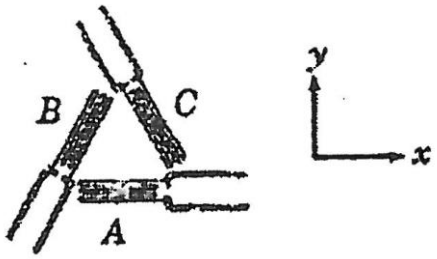
解得

$$\varepsilon_y = 200.3 \times 10^{-6} \qquad \gamma_{xy} = 1559.2 \times 10^{-6}$$

$$\sigma_x = \frac{E}{1-\nu^2}(\varepsilon_x + \nu\varepsilon_y) = 91.6 \text{ MPa}$$

例 36

如圖所示之正三角形應變計組合，黏貼於鋼材之表面。已知圖中的三個應變計讀數分別爲 $\varepsilon_A = 374\times10^{-6}$，$\varepsilon_B = -135\times10^{-6}$，$\varepsilon_C = 227\times10^{-6}$，且鋼之 E = 210 GPa、柏松比 $\nu = 0.3$。試求主應變 (principal strains)之大小及方向。



【中央土木】

【解析】

$$\varepsilon_x = \varepsilon_A = 374\times10^{-6}$$

$$\varepsilon_y = \frac{2}{3}(\varepsilon_B + \varepsilon_C) - \frac{1}{3}\varepsilon_A = -63.33\times10^{-6}$$

$$\gamma_{xy} = \frac{2\sqrt{3}}{3}(\varepsilon_B - \varepsilon_C) = -418\times10^{-6}$$

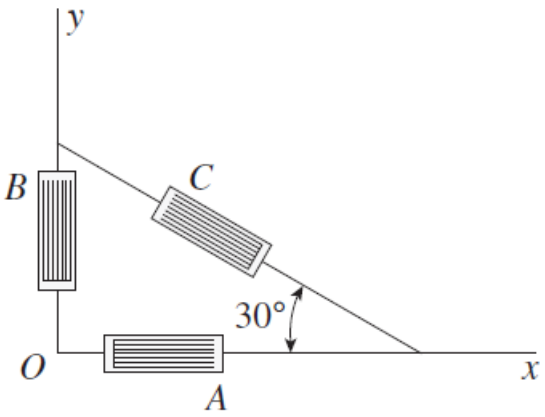
$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{(\frac{\varepsilon_x - \varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2}$$

$$= 457.82\times10^{-6}, -147.14\times10^{-6}$$

$$\theta_P = \frac{1}{2}\tan^{-1}(\frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y}) = -21.85^\circ, 68.15^\circ$$

例 37

On the surface of a structural component in a space vehicle, the strains are monitored by means of three strain gages arranged as shown in the figure. During a certain maneuver, the following strains were recorded: $\varepsilon_a = 1100 \times 10^{-6}$, $\varepsilon_b = 200 \times 10^{-6}$, $\varepsilon_c = 200 \times 10^{-6}$Determine the principal strains and principal stresses in the material, which is a magnesium alloy for which $E = 40\,\text{GPa}$ 及 $\nu = 0.35$. (Show the principal strains and principal stresses onsketches of properly oriented elements.)



【102 交大土木類題】

【中文翻譯】

太空船結構之元件表面上，應變由三個應力規所監視，其安排如圖所示。在太空船做了某運動時，記錄了如下之應變： $\varepsilon_a = 1100 \times 10^{-6}$, $\varepsilon_b =$ 200×10^{-6} 及 $\varepsilon_c = 200 \times 10^{-6}$ 。求材料之主應變及主應力；材料為鎂合金，具 $E = 40\,\text{GPa}$ 及 $\nu = 0.35$ 。在正確指向的元素上繪示主應變及主應力。

【解析】

$\theta = 0°$ $\quad \varepsilon_A = 1100 \times 10^{-6}$

$\theta = 90°$ $\quad \varepsilon_B = 200 \times 10^{-6}$

$\theta = 150°$ $\quad \varepsilon_C = 200 \times 10^{-6}$

$$\varepsilon_x = \varepsilon_A = 1100 \times 10^{-6}$$

$$\varepsilon_y = \varepsilon_B = 200 \times 10^{-6}$$

$\theta = 150°$:

$$\varepsilon_{x_1} = \varepsilon_C = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

$$200 \times 10^{-6} = 650 \times 10^{-6} + 225 \times 10^{-6} - 0.43301\gamma_{xy}$$

$$\gamma_{xy} = 1558.9 \times 10^{-6}$$

主應變

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

$$= 650 \times 10^{-6} \pm 900 \times 10^{-6}$$

$$\varepsilon_1 = 1550 \times 10^{-6} \qquad \varepsilon_2 = -250 \times 10^{-6}$$

$$\tan 2\theta_p = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} = \sqrt{3} = 1.7321$$

$$2\theta_p = 60° \qquad \theta_p = 30°$$

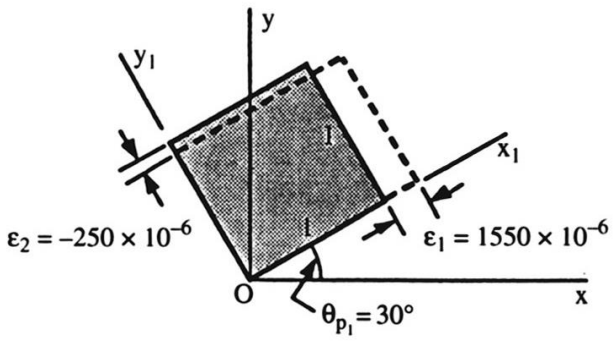
$\theta_p = 30°$:

$$\varepsilon_{x_1} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

$$= 1550 \times 10^{-6}$$

$$\therefore \ \theta_{p_1} = 30° \qquad \varepsilon_1 = 1550 \times 10^{-6}$$

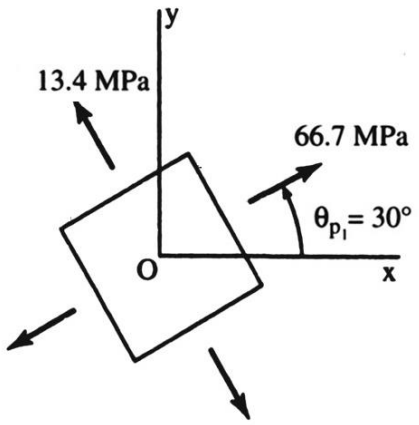
$$\theta_{p_2} = 120° \qquad \varepsilon_2 = -250 \times 10^{-6}$$



)

$$\sigma_1 = \frac{E}{1-\nu^2}(\varepsilon_1 + \nu\varepsilon_2), \ \sigma_2 = \frac{E}{1-\nu^2}(\varepsilon_2 + \nu\varepsilon_1)$$

$$\sigma_1 = 66.7 \text{ MPa}, \ \sigma_2 = 13.4 \text{ MPa}$$


y
13.4 MPa
66.7 MPa
θ_{p_1}= 30°
O
x

例 38

A strain rosette (Fig.2), composed of three resistance gages making angles of 0^0, 60^0, and 120^0 with the x-axis, was used to measure the strains on the free surface of a steel component (Young's modulus = 3×10^7 psi and Poisson's ratio = 0.3). The measured strains were $\varepsilon_0 = \varepsilon_x = +10^{-3}$, $\varepsilon_{60} = -0.65\times10^{-3}$, $\varepsilon_{120} = +0.75\times10^{-3}$. (a) Determine the principal strains and the maximum shearing strain at the point (10%); (b) show the directions of the in-plane principal strains on a sketch (5%); (c) determine the principal stresses and the maximum shearing stress at the point (10%); (d) show the principal stresses and the maximum shearing stress on a sketch (5%).

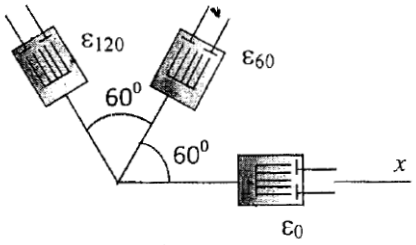


Fig. 2

【104 成大航太】

【解析】

(一)

$$\varepsilon_x = 1000\,\mu = \varepsilon_0$$

$$\varepsilon_{120^\circ} = \frac{\varepsilon_x+\varepsilon_y}{2} + \frac{\varepsilon_x-\varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = 750\mu$$

$$\Rightarrow \frac{1000+\varepsilon_y}{2} + \frac{1000-\varepsilon_y}{2}\cos 2\times(120^\circ) + \frac{\gamma_{xy}}{2}\sin 2\times(120^\circ) = 750\mu$$

同理

$$\varepsilon_{60^\circ} = \varepsilon_c = \frac{\varepsilon_x+\varepsilon_y}{2} + \frac{\varepsilon_x-\varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = -650\mu$$

$$\Rightarrow \frac{1000+\varepsilon_y}{2} + \frac{1000-\varepsilon_y}{2}\cos 2\times(60^\circ) + \frac{\gamma_{xy}}{2}\sin 2\times(60^\circ) = -650\mu$$

由上兩式可解得

$$\varepsilon_y = -\frac{800}{3}\mu \text{ 、 } \gamma_{xy} = -\frac{2800}{\sqrt{3}}\mu$$

(二)

$$\frac{\gamma_{\max}}{2} = \sqrt{(\frac{\varepsilon_x-\varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2} \Rightarrow \gamma_{\max} = 2053.7\mu$$

$$\varepsilon_{1,2} = \frac{\varepsilon_x+\varepsilon_y}{2} \pm \sqrt{(\frac{\varepsilon_x-\varepsilon_y}{2})^2 + (\frac{\gamma_{xy}}{2})^2} = 1393.5\mu, -660.2\mu$$

由虎克定律

$$\varepsilon_1 = \frac{1}{E}(\sigma_1 - \upsilon\sigma_2) \Rightarrow 1393.5\mu = \frac{1}{3\times10^7}(\sigma_1 - 0.3\sigma_2)$$

$$\varepsilon_2 = \frac{1}{E}(\sigma_2 - \upsilon\sigma_1) \Rightarrow -660.2\mu = \frac{1}{3\times10^7}(\sigma_2 - 0.3\sigma_1)$$

$\sigma_1 = 39.41(ksi), \sigma_2 = -7.98(ksi)$

$\tau_{\max} = \dfrac{\sigma_1 - \sigma_2}{2} = 22.7(ksi)$

